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# iGEM deliverables plan: wiki deadline, depolymerase assays, and sequencing

[00:00:00] Speaker 1: Uh, Wednesday.

[00:00:01] Speaker 2: Wednesday? Sounds good, sounds good. All right, I see.

[00:00:05] Speaker 1: Awesome.

[00:00:06] Speaker 2: So, yeah, there are a few places we can start right now. So, first of all, we can doubly walk through, like, what are required by item, and then Olivia can talk about that. And there is another thing I want to talk about today, which, which is, like, what are the asset or, like, figures or, like, things we can talk about right now. So that's the second thing I want to talk about.

[00:00:29] Speaker 1: What do you mean, uh, as to talk about what, exactly?

[00:00:32] Speaker 2: It's like, so, like, what can we show? It's like, our current results.

[00:00:36] Speaker 1: Okay, yeah, yeah, no, yeah, I get it. I understand, yeah, yeah.

[00:00:40] Speaker 2: Mm-hmm.

[00:00:40] Speaker 1: Yeah.

[00:00:43] Speaker 2: Ugh, sounds good. So where should.

[00:00:45] Speaker 1: Yeah, yeah, yeah.

[00:00:46] Speaker 2: Where should we shop? Where should we— where should—

[00:00:47] Speaker 1: Basically the paper. That would be the paper. But don't we start by the requirements?

[00:00:52] Speaker 2: Yeah, sure.

[00:00:53] Speaker 1: And then I tell you what the idea—

[00:00:55] Speaker 2: Mm-hmm.

[00:00:55] Speaker 1: The idea that I have for the paper, and also that that's, like, frames nicely, like, what's the, um, the, the stuff that you guys have, right?

[00:01:04] Speaker 2: Mm-hmm. Sounds good.

[00:01:07] Speaker 1: Okay. So fireway, yes.

[00:01:12] Speaker 3: So should I start with the deliverables?

[00:01:15] Speaker 1: Mm-hmm. Like, if you have a sh— you have a screen to share or something that—

[00:01:19] Speaker 3: Yeah.

[00:01:19] Speaker 1: That does help.

[00:01:20] Speaker 3: Um, I actually— wait, let me check. Because I just, like, I didn't have a, like, a deadline calendar or something. I just exported to, like, my Apple calendar. But I, I can open the official website, I think.

[00:01:36] Speaker 2: Mm-hmm.

[00:01:37] Speaker 3: Let me check. Um, okay, probably this one. Can you see my screen right now?

[00:01:44] Speaker 1: Uh, yes, yes. I see— we see the, uh, iGen, uh, page.

[00:01:48] Speaker 3: Yeah, yeah, yeah. Okay, so, okay, let me check. Deliverables. And—

[00:01:59] Speaker 1: Mm-hmm.

[00:01:59] Speaker 3: Wait, let me close this one so I can—

[00:02:01] Speaker 2: Okay.

[00:02:02] Speaker 3: Okay, so, like, here's like where we are right now.

[00:02:06] Speaker 1: Okay, okay.

[00:02:07] Speaker 3: And, like, oh, why don't I just— I'll just share my calendar, I think, because, like, that's, like, more clear, I guess. Wait a second.

[00:02:16] Speaker 1: Like, you have a calendar? Yes.

[00:02:18] Speaker 3: Yeah, I think the calendar is better. Hmm, printing something else. Entire screen. I'll just— oh my god. Um, um, oh my god. And, okay, I'll just go with this website, I think.

[00:02:31] Speaker 1: Yeah.

[00:02:31] Speaker 3: So, like, our, um—

[00:02:34] Speaker 1: That's it.

[00:02:34] Speaker 3: Like, the—

[00:02:35] Speaker 1: I guess—

[00:02:35] Speaker 3: Huh?

[00:02:36] Speaker 1: No, go, go for it. Go for it, yes.

[00:02:38] Speaker 3: Oh yeah. Our Jamboree stuff just, like, came out, like our, like, Q&A session and also the live street talk. And, like, that's, like, something we need to— we'll do in Paris. And speaking of the deliverables, um, there's a, um, team photo and team logo deal on Wednesday, and I'm coordinating with everyone about the time so we can gather together to get a group photo. And—

[00:03:04] Speaker 1: Oh, wait, yeah, yeah, yeah. Let me know that.

[00:03:07] Speaker 3: Yeah. And that one is, like—

[00:03:08] Speaker 1: Just text me, and then send it to me so we can put it on my website.

[00:03:11] Speaker 3: Yeah. And that one is, like, something to put right over here, like, the group photo.

[00:03:15] Speaker 1: Okay.

[00:03:16] Speaker 3: And, like, our, like, I plan, uh, the photo to be taken on Wednesday at 4:30 PM. And the people I want to gather will be, like, the person who are doing the experiment. So.

[00:03:31] Speaker 1: Yes.

[00:03:32] Speaker 3: We asked Carol and Maggie, also Rafael, and I, I also pro.

[00:03:37] Speaker 1: I don't need Rafael. I don't need Rafael. I mean, you can include Rafael, but I'm concerned that Rafael is a grad student, right?

[00:03:43] Speaker 3: Uh, it's fine. It's fine. I asked him.

[00:03:45] Speaker 1: Okay.

[00:03:46] Speaker 3: That they have—

[00:03:47] Speaker 1: Let's include Rafael.

[00:03:48] Speaker 3: Yeah. And probably we'll also ask Waitell to come that day because he'll be the human practitioner.

[00:03:55] Speaker 1: Yeah, no, you guys have, like, free, like.

[00:03:57] Speaker 3: Yeah, yeah, yeah. Can you come on Wednesday, bro?

[00:04:00] Speaker 1: Like, if you want me to be in the picture, yes, I can be there.

[00:04:02] Speaker 3: Uh, sure. Yeah. Because I, I want, like, people to wear our uniform and also the lab coat so we can have a picture together.

[00:04:09] Speaker 1: Oh, I see. Yeah, yeah, yeah. Okay, yeah.

[00:04:11] Speaker 2: By the way, prof.

[00:04:12] Speaker 3: And that's, like—

[00:04:14] Speaker 2: Yeah, we have a favor to ask. Can we— should we add Rafael as a part of the, like, advisor to the team? Because there is a— there is a position to.

[00:04:25] Speaker 3: Oh, no.

[00:04:25] Speaker 2: Yeah.

[00:04:25] Speaker 3: No, no, no, no. The, the, the roster is already due, so we cannot add more people to this roster. We can only keep somebody else.

[00:04:31] Speaker 2: Oh, yeah. We can— can we keep.

[00:04:33] Speaker 3: Yeah. But, like, speaking of, like, there's gonna be something called attribution, which means, like, which one, like, which people is in charge of which kind of stuff.

[00:04:41] Speaker 1: Oh, which part, yeah, yeah.

[00:04:42] Speaker 3: And, like, yeah, we can put, like, other resources, like people who are not on the roster.

[00:04:47] Speaker 1: Yes.

[00:04:47] Speaker 3: And we can put, like, put Rafael there. And probably also, like, uh, Botanic Garden or, like, any resource that we reach out to. And we can just put it on our wiki but not in the roster.

[00:04:58] Speaker 1: Yeah, yeah. I mean, like, many, many big labs. If we had a big lab, we would have lab techs, right?

[00:05:03] Speaker 3: Yeah.

[00:05:03] Speaker 1: And a lab tech would not be part of the team, right? Because that's their job, right?

[00:05:06] Speaker 3: Yeah.

[00:05:06] Speaker 1: So if you— Rafael could be something like that.

[00:05:09] Speaker 3: Yeah, yeah, yeah, yeah. It's just, like, kind of, like, because I didn't check the deadline of.

[00:05:15] Speaker 1: No, let's not think about it.

[00:05:16] Speaker 3: Okay.

[00:05:16] Speaker 1: I, my only time I thought Rafael was gonna be not in the team because also, like, that bumps you to the next class, right?

[00:05:23] Speaker 3: Yeah, yeah. We just, like, I, I didn't— I, I didn't check the deadline, so I forgot to add, like, there are also some other people who's not on the roster, like Maggie, and she's not on the roster. But, like, we can just, like.

[00:05:35] Speaker 1: That's true.

[00:05:36] Speaker 3: Keep their attribution.

[00:05:36] Speaker 1: And, and Phoebe. I don't know if Phoebe's still there, you know, and stuff like that.

[00:05:40] Speaker 3: Uh-huh.

[00:05:40] Speaker 1: And we need to, uh, yeah, I mean, they've— I mean, for sure, I mean, Rafael will get credited on the paper papers, right?

[00:05:46] Speaker 3: Mm-hmm.

[00:05:47] Speaker 1: But, like, yeah, so this is— this is— this is a completely different thing, so.

[00:05:50] Speaker 3: Uh, yeah. Yeah, it's just, like, something about the competition itself. Yeah. And other deadlines is that for the wiki, it's gonna be a harsh deadline, which is October 21st, and there's, like, a lot of stuff due that day. And for the wiki, I already just called, uh, let me check, share this tab instead. Can you see the wiki? I'm just, like, building the.

[00:06:17] Speaker 1: Mm-hmm.

[00:06:18] Speaker 3: The framework. I, like, I borrowed the idea from, like, some, like, like, winner team about, like, how their, like, direct, uh, tab it looks like. And, like, for each of them, I just asked Claude to do the structure. And we can, like, uh, fill in any text or pictures we want later.

[00:06:38] Speaker 1: Yeah.

[00:06:39] Speaker 3: And also we can change the design by just, like, a framework.

[00:06:43] Speaker 1: Yeah. Yeah, no, I, I like it. I like it.

[00:06:45] Speaker 3: Uh.

[00:06:45] Speaker 1: I mean, I mean, this is all aesthetics, right?

[00:06:48] Speaker 3: Yeah.

[00:06:48] Speaker 1: So maybe, maybe we can a.

[00:06:51] Speaker 3: Yeah.

[00:06:51] Speaker 1: We can definitely— we should definitely concentrate on the results, and then, like, build from that.

[00:06:55] Speaker 3: Yeah.

[00:06:55] Speaker 1: Or, I mean, like, whatever aesthetic.

[00:06:58] Speaker 3: Mm-hmm.

[00:06:59] Speaker 1: Appearance you choose, uh, it's fine.

[00:07:01] Speaker 3: Yeah.

[00:07:01] Speaker 1: The only problem, like, how with this one.

[00:07:03] Speaker 3: Mm-hmm.

[00:07:04] Speaker 1: Is just, like, it looks straight from, uh, from cloth.

[00:07:10] Speaker 3: Yeah.

[00:07:10] Speaker 2: Yeah, we can fix.

[00:07:11] Speaker 1: Yeah.

[00:07:11] Speaker 2: We can fix the aesthetic later. We can definitely fix the aesthetic later.

[00:07:13] Speaker 1: I know, I know, I know. I'm just— exactly what I'm saying. Like.

[00:07:16] Speaker 3: Yeah.

[00:07:16] Speaker 1: Uh, the aesthetic is the least important thing right now.

[00:07:19] Speaker 3: Yeah.

[00:07:19] Speaker 1: We should focus on the results. But I'm saying, like, the aesthetic in this is perfect. It's just, like, my only concern is, like, it looks straight from cloth, right?

[00:07:27] Speaker 3: Uh-huh.

[00:07:27] Speaker 1: And, yeah, so there's something that we can definitely add.

[00:07:30] Speaker 3: Like, it's something that's, like, easier to change later.

[00:07:33] Speaker 1: Yeah, yeah.

[00:07:33] Speaker 3: And the, the, the top, like, the, the higher priority is, like, just, just, like, get the everything's, like, our description were, like.

[00:07:42] Speaker 1: I feel like you should actually start— let's start with a graphical abstract, actually.

[00:07:45] Speaker 3: Yeah.

[00:07:45] Speaker 1: I think if you start with a graphical abstract.

[00:07:47] Speaker 3: Uh-huh.

[00:07:48] Speaker 1: You'll immediately know what you're gonna talk about, right?

[00:07:50] Speaker 3: Yeah.

[00:07:50] Speaker 1: And I think, again, it's, it's, it's the paper, right?

[00:07:52] Speaker 3: Mm-hmm.

[00:07:52] Speaker 1: It's the paper.

[00:07:53] Speaker 3: Yeah. Yeah, also, like, the human practice. I already sent the email to Ashley, and I'm waiting for her response.

[00:08:00] Speaker 1: Yeah, yeah. Hopefully she will reply soon.

[00:08:02] Speaker 3: Yeah. And since, like, the, the, the deadline of, of the, the wiki is 21st, which is, like, during the, uh, during our fall break, we need to get the interview done by then. And Alex and I will work on the wiki and also other, like, deliverables.

[00:08:20] Speaker 1: You guys have a fall break? You guys have a fall break?

[00:08:22] Speaker 3: Yeah, we have a fall break next, like, from, I think it's 17th or, like, the next Thursday. No, next Saturday to, like, Tuesday.

[00:08:32] Speaker 1: Okay, so it's a short break.

[00:08:34] Speaker 3: Yeah. It's a very short break. So we're just gonna.

[00:08:36] Speaker 1: Yeah.

[00:08:36] Speaker 3: Focusing on the weekend and also other things.

[00:08:39] Speaker 1: Yeah.

[00:08:40] Speaker 3: Yeah. And one week after that, there's a presentation video due, which is, like, basically how is your project doing right now. And it's, like, quite easy.

[00:08:52] Speaker 1: Yeah.

[00:08:53] Speaker 3: It doesn't have to be fancy. You can just, like, make a slide and just, like, share your screen on Zoom and record the meeting, and you can just, like, present your stuff. And it's, like, pretty easy as long as we can, like, summarize how we are doing so far.

[00:09:08] Speaker 1: Mm-hmm. I like it. I like it.

[00:09:09] Speaker 3: Yeah. Uh, and that's, like, kind of similar to the judging session because, like, for the judging session, we also have a, I think it's five-minute, like, five-minute introduction about our project. And then for the rest, I don't know if 20 or 15 minutes, it's gonna be Q&A. So it's, like, a, a overview for the judges to know what, what your project is about.

[00:09:31] Speaker 1: Yeah.

[00:09:31] Speaker 3: So it's, like.

[00:09:32] Speaker 1: So the presentation video is five minutes, and it's just mostly an introduction of the paper, of the project.

[00:09:39] Speaker 3: Uh.

[00:09:39] Speaker 2: Presentation video is, like, 10 minutes. It's, like, 10 minutes to 15 minutes.

[00:09:43] Speaker 3: No, it, it, it's 15.

[00:09:45] Speaker 2: It's 15 minutes.

[00:09:45] Speaker 1: But it shows that it shows it has, like, it's like a, it's like a research talk.

[00:09:49] Speaker 2: Yeah, it's like a research talk.

[00:09:50] Speaker 1: It's called okay, okay.

[00:09:52] Speaker 2: And it's more like.

[00:09:53] Speaker 1: It's just like. It's just like a question. Yeah.

[00:09:55] Speaker 3: Uh-huh.

[00:09:55] Speaker 2: So it's, like, combining the whole year's research into, like, one 15-minute video. So, like, basically, we'll talk about stuff like how you do, like, interview, including, like, how you do interview, like, how you do the experiment, what happened, what did you find out. So it's more like a whole journey thing instead of, like, just the lab work.

[00:10:13] Speaker 1: Uh-huh.

[00:10:13] Speaker 2: So it's, like, broader than just the research.

[00:10:16] Speaker 1: Awesome.

[00:10:16] Speaker 2: Yeah.

[00:10:17] Speaker 1: Yeah, yeah.

[00:10:18] Speaker 3: Uh, could you clarify on the, like, Q&A session? Is that also in the pre, uh, presentation video?

[00:10:25] Speaker 1: No, no, no, no.

[00:10:25] Speaker 2: That's in judging session.

[00:10:26] Speaker 3: It's part of the real judging, which happens in November. And that's online or, like, in. Uh, it can be hybrid. Like, it can be Alex and I, uh, doing the introduction. Like, also, like, there's gonna be a Zoom meeting link, I think. Like.

[00:10:40] Speaker 1: Oh, very nice.

[00:10:40] Speaker 3: Because for the judges, some of them are also online. So you can just join the, uh, meeting room. And, like, for the judging session, like, for people staying, uh, online, they can also answer the question. But, like, Alex and I can do the, like, five-minute introduction for the judging session.

[00:10:55] Speaker 4: So it's like we upload our presentation video to this, um, website.

[00:11:00] Speaker 3: Uh, like, the presentation video is, like, something submitted before the Jamboree. So, like, you're, like, you're gonna be assigned to, like, several judges, and they will take a look at your, uh, presentation video and also the wiki beforehand. And during the judging session, which is, like, in the mid of November, they're just gonna ask you about, like, whatever they wanna ask, like, what's their concern, like, any question. It's just gonna be based on the presentation video and the wiki.

[00:11:29] Speaker 4: Got it. Got it.

[00:11:30] Speaker 3: Yeah. Yeah. And, yeah, that's, like.

[00:11:35] Speaker 2: And, like, for other.

[00:11:36] Speaker 3: Judging.

[00:11:37] Speaker 2: For other, like, deliverables, something like judging form, we just have to, like, submit the specific links from our wiki to the page. So, like, for example, like, which part is the engineering success, which part is, like.

[00:11:49] Speaker 1: Mm-hmm.

[00:11:49] Speaker 2: Project description. It's more like a, more like a directory for them to, like, figure out, like, what do we do. So that's more on the judging form.

[00:11:57] Speaker 3: Okay.

[00:11:57] Speaker 2: For software, I'm not sure we do you have to submit any software.

[00:12:02] Speaker 3: We don't have a software.

[00:12:02] Speaker 2: Because.

[00:12:03] Speaker 1: I think we do have software, and I should send you this.

[00:12:05] Speaker 2: Yes.

[00:12:05] Speaker 1: What does that do?

[00:12:07] Speaker 2: So that is also due the day wiki is due. So.

[00:12:09] Speaker 3: 21st.

[00:12:10] Speaker 2: For the software part.

[00:12:11] Speaker 1: Yeah.

[00:12:13] Speaker 2: Here's what I think.

[00:12:14] Speaker 1: I'll, I'll yeah, yeah, go ahead. Go ahead.

[00:12:16] Speaker 2: Yeah.

[00:12:16] Speaker 1: I was gonna let me finish just.

[00:12:18] Speaker 2: Mm-hmm.

[00:12:18] Speaker 1: Because I do have software, and I think for software, we can do, uh, it is I don't know. Software would be a good way to call it. It's just, like, uh, the, uh, phylogenetic, uh, inference, right? Like, ancestral stereo constr I did ancestral stereo construction, and I also, uh, fitted the fake proteins to a, to a language model that'll tell me how well they fit to a particular carboxylase.

[00:12:48] Speaker 2: Yes. I think for, for that, what we can do is that we can, we can have a page about, like, all the tools we have. So, like, here's, like, for example, like, AlphaFold. Here's, like, the, the large language model for the phylogenetic tree. Here's a part for constructing a plasmid. We can have all these different tools lined up. That can be one thing. The other thing is that.

[00:13:09] Speaker 1: Mm-hmm.

[00:13:09] Speaker 2: If we have time, we can try to, like, connect them into, like, a pipeline or something like that.

[00:13:15] Speaker 1: Yes.

[00:13:15] Speaker 2: So that we can, like.

[00:13:16] Speaker 1: That's best.

[00:13:17] Speaker 2: Submit.

[00:13:17] Speaker 1: That's best. Yes.

[00:13:18] Speaker 2: Submit into a into software as, like, one single thing. So I think that would be better.

[00:13:22] Speaker 1: That's what I was saying. Like, you can have, like, different tiers. Like, the minimum tier is just a list of software.

[00:13:26] Speaker 2: Yeah.

[00:13:27] Speaker 1: And a, uh, uh, uh, polished stuff would be a pipeline.

[00:13:32] Speaker 2: Yeah. Yeah. Something like that.

[00:13:34] Speaker 1: Mm-hmm.

[00:13:36] Speaker 2: So. And then for other deliverables, something like attribution form, it's basically just, like, who did who like, who did what.

[00:13:43] Speaker 1: Yeah.

[00:13:43] Speaker 2: That's, that's, that's basically it. That's, that should be pretty straightforward. And then one last thing. Did so I need to make sure did Jasmine or, like, anyone from KJI reply to that application?

[00:13:55] Speaker 1: I, I emailed I mean, she replied last week. It's just, like, she was telling me it's like, this is, like, peer review kind of thing.

[00:14:02] Speaker 2: I see.

[00:14:03] Speaker 1: Like, she has to gather enough people.

[00:14:04] Speaker 2: Uh-huh.

[00:14:05] Speaker 1: So they can look at my petition form.

[00:14:07] Speaker 2: Because safety form is due in two days. So if there's, like, if there's no, like, certification, can you go up there and just, like, like,

submit and then, and then, like, reply to the feedback? Because, like, I tried to make all the all, like, all the how can I say? All the adjustment I did last time. And then after that, can you, like, make a final check and say, like, "Hey, it's, like, our final safety form"?

[00:14:33] Speaker 1: Okay. Okay. Yeah, I can do that. I can do that.

[00:14:35] Speaker 2: Yeah.

[00:14:35] Speaker 1: I can do that. Yeah. Awesome. Yeah. Let me, um, can you show me where it is? And I'll, I'll just.

[00:14:44] Speaker 2: Yes. Uh, you can go into your go into your account. There's a.

[00:14:48] Speaker 1: Yes.

[00:14:49] Speaker 2: Yeah. I'll, I'll, I'll leave you to introduce it.

[00:14:51] Speaker 3: Can you start from, from the first, like, this is the homepage.

[00:14:53] Speaker 1: Yes. Uh-huh.

[00:14:54] Speaker 3: And then you log in.

[00:14:56] Speaker 1: Yes.

[00:14:57] Speaker 3: And you click the, the, the nine dots, like, uh, click into Teams.

[00:15:02] Speaker 1: Uh-huh.

[00:15:02] Speaker 3: And then you can find our team under your teams, like Clemer CA.

[00:15:07] Speaker 1: Yes.

[00:15:08] Speaker 3: And then you scroll down. There's a, uh, tab called Deliverables. And then you'll scroll down. There's a project safety form here. And you can just.

[00:15:21] Speaker 1: Okay.

[00:15:22] Speaker 3: Click.

[00:15:22] Speaker 1: So it's just a project safety form. Okay.

[00:15:24] Speaker 2: Mm-hmm.

[00:15:25] Speaker 3: Yep.

[00:15:25] Speaker 2: Mm-hmm.

[00:15:25] Speaker 3: Yes.

[00:15:28] Speaker 2: Yeah. Um.

[00:15:29] Speaker 1: Got it.

[00:15:29] Speaker 2: A bit more update about, like, from my side, other than the deliverables, I'm talking to our president from, like, the new president from CMC on October 23rd. And we'll see is there any update to the all the stuff. Yeah.

[00:15:44] Speaker 1: Yeah. No, honestly, to meet with them, why don't you actually I think it will help you very much if you come up with, like, the graphical abstract.

[00:15:52] Speaker 2: Mm-hmm.

[00:15:54] Speaker 1: And the paper that's already published.

[00:15:56] Speaker 2: Oh, yeah. Uh, do you want to talk about paper now? Because I think that's, that's most of the part for the for the deliverables, right, Olivia?

[00:16:03] Speaker 1: Yeah, yeah. No, I yeah, the paper, I thought it's actually very it's summarizing the figure that I sent you guys, right?

[00:16:09] Speaker 2: Mm-hmm. Mm-hmm.

[00:16:10] Speaker 1: Uh, I guess it's not a great figure, but let me, uh so the idea is, uh, I.

[00:16:19] Speaker 2: Probably do you want to share your screen?

[00:16:22] Speaker 1: Yeah. So let me, uh what I think, uh let me, let me gather my figures. Let me gather my, uh, the figures that I think could be good for your, uh the, the paper. I wanna sh let's talk about the paper first. And then from that, I think you can use all of these figures for your, uh, deliverables there. And on top of that, you can, uh, like, uh, add extra stuff, you know? I think, like, for example, the genomes, right? The genomes, um, of all of your, uh, the things that we isolated, uh, protein, uh, 3D models, etc., etc. So the idea of hold on. So, okay. Just stop. Just give me one second. Okay. Okay. Research and pathology. Okay.

[00:17:33] Speaker 2: What journal are you are you looking forward to, like, submit to?

[00:17:38] Speaker 1: No, no, I'm looking for, like, some, uh, some of the stuff that you wrote, right?

[00:17:41] Speaker 2: Mm-hmm. Yeah. I mean, like, what, like, which journal are you looking forward to, like, submit to?

[00:17:47] Speaker 1: Oh, sorry. I think I mean, it since it's just my genome and stuff like that, I'm thinking microbiology spectrum.

[00:17:54] Speaker 2: Hmm. I see.

[00:17:55] Speaker 1: Uh-huh. Uh, or MC stems, or, uh, uh, something like that. Maybe we really we could try, like, uh, ACM synthetic biology because we can say, like, "Oh, those are, like, fake proteins that we, uh, created," right?

[00:18:12] Speaker 2: Mm-hmm. Mm-hmm.

[00:18:13] Speaker 1: Uh, stuff like that. But I think, like, at minimum, it's a microbiology spectrum, right, which is, is not bad, you know? I think a idea, like, if we my, my, my grand plan is, like, to get, like, a ton of fish. So we could if we have a ton of fish, it could honestly be a nature microbiology and nature EEV, or a nature microbiology or communications biology, like that. But for now, we need a lot more fish, right?

[00:18:42] Speaker 2: Right.

[00:18:43] Speaker 1: Just, just one fish. So there's that. So, uh, so I'll, I'll share my screen really quickly. And I can probably send you this document as well. So, uh, window, entire screen. Awesome. Share. Okay. So you can see this thing, right? So, uh, for instance, this is, uh, this is, uh can you see it?

[00:19:05] Speaker 3: Yes.

[00:19:05] Speaker 1: Yes. Awesome. So this is what I think could potentially be the, the idea of the paper, right? So basically, much answer, uh, uh, answers how, like, how the species are int like, how things get, uh, explored into novel, uh, uh, niches, right? Again, like, question is, whose range is central traits of predatory lineages, right? But we the study is heavily biased, right? Most of this is based on binding receptor binding proteins or some, uh, post-entry defenses, like CRISPR, like modifi uh, restriction modification, uh, uh, systems, right? And both of them act at the absorption, right? In many bacteria, there's a filter that is beforewrite, right? And does the capsule must be chopped before the receptor binding protein can be engaged with the receptor of the.

[00:20:01] Speaker 2: Mm-hmm.

[00:20:02] Speaker 1: The, of the cell, right?

[00:20:03] Speaker 2: Mm-hmm.

[00:20:04] Speaker 1: And so fish depleting rays perform that degrade degradation, right? We know they're widespread, and they are poorly characterized biochemically. And, and more importantly, we don't really know what's the that's the evolutionary question, right? Uh, and then linking the relationship between the host range and, uh, uh, has not been characterized biochemically, right? So our hypothesis is that the depleting rays, uh, like, the, the breadth of substrate sets some sort of an independent range of, uh, hosts, right? And that many maybe the narrow specificity that we see in present day enzymes is derived from broader ancestral states.

[00:20:49] Speaker 2: Hmm.

[00:20:50] Speaker 1: So that's the hypothesis, right? So this reframes, like, host range as a trait with its own evolutionary history, right? And makes the, uh, uh and makes this, uh, generalist especially question, which is being studying in plants, insects, something that we can actually do in the lab really easily, right? Because we can resurrect ancestral states, right? Uh, so we have we established our system, right? We have isolated phages, phage, phages, at least two of them. Maybe they are the same. We hope to isolate more, right?

[00:21:27] Speaker 2: Mm-hmm.

[00:21:28] Speaker 1: And we'll look at the candidate polymerase sequence, uh, distinct varying positions, right? And these are the these are the polymerase states, right? So we have phage, we have the genomes. We know some of them are the depolymerases. This is a cartoon, right? Hopefully, we're gonna get more, uh, more of this, right? Uh, with your help, we'll help other students, right? We know what we have is what we have, right? Uh, basically, just, uh, try and or isolate, uh, or polymerase.

[00:21:57] Speaker 2: Mm-hmm.

[00:21:58] Speaker 1: Against all their, uh, pseudomonas, right? Do they kill it? Yes or no, right?

[00:22:01] Speaker 2: Right.

[00:22:03] Speaker 1: So and then what we're gonna what we did is, uh, we're gonna basically, what we're trying to do is, is this right, right? Uh, so here's we're gonna extend this is, like, a this is for a plan. This is for a, a job application. Uh, extend, uh we're gonna extend the, uh we're gonna extend and test the evolutionary history of the specificity of these depolymerases, right? First, we're gonna do metagenomes. We couldn't do it, but, like, that's future plan. I mean, at the end, with the end of the day, 'cause we use, like, published sequences from Magnify. Hopefully, you remember, right?

[00:22:38] Speaker 2: Yep. Mm-hmm.

[00:22:38] Speaker 1: Basically, just create a huge, uh, tree, phylogenetic tree, and we reconstruct the ancestral states, right? Uh, and basically, we wanna characterize the phenotype of these and now, we're characterizing the phenotype of these current and ancestral enzymes, right? Uh, and we, uh, we ex we are expressing them. We're characterizing cha biochemically, right? You're there already. We know you can make proteins already, right? We have some of them, at least one of them, right? And the idea is we wanna test, like, if the the idea is, like, we think that, uh, deeper notes have broader, uh, uh, broader substrate range than, than the old ones, right? And we're gonna and we're gonna relate this, uh, uh, uh, specific this biochemical, uh we're gonna relate this biochemical, uh, results to the actual host range of phages, right? So basically, once we have this, we're gonna test if this that is the host range is recapitulated in somehow by this, uh, by the biochemistry of enzymes, right?

[00:23:55] Speaker 2: Mm-hmm. Right.

[00:23:56] Speaker 1: And one of the most experiments that I'm the more excited about is, is this one, right? A key experiment is gonna be we're gonna

supply the purified enzyme in trance, and we're gonna see if it extends the range of receptor binding protein, right?

[00:24:11] Speaker 2: Mm-hmm.

[00:24:11] Speaker 1: That is pretty much like we talked about, right? Like, if we add my enzyme, can this host now be accessible to other phages, right? And that's the and that's the idea here.

[00:24:26] Speaker 2: Mm-hmm.

[00:24:26] Speaker 1: So that's to me, that's, that's the that's the mini-paper, right? So that is basically saying, like, uh, we're asking the question, is the, uh is the host range independently codified by depolymerases, right? Does all of these uh, well, well, quick question. All of these questions are very basic scientifically, right? But they are, like, applications, right? They could potentially be translated, right? Because maybe we can engineer these, so they so it ha so it is it extends host range, right?

[00:24:54] Speaker 2: Mm-hmm.

[00:24:54] Speaker 1: Like, ultimately. Again, like, the questions right now are, like, pure scientifically, but you can always start saying, like, "Okay." And ultimately, maybe we can engineer these, uh.

[00:25:04] Speaker 2: Phages.

[00:25:05] Speaker 1: We can engineer host range, right?

[00:25:06] Speaker 2: Yep.

[00:25:06] Speaker 1: 'Cause we know that, uh, there is, uh, the, the host range is at least partially codified by the depolymerase, which is this one, not just this one, right?

[00:25:18] Speaker 2: Mm-hmm.

[00:25:19] Speaker 1: And that's it. So that's what I think the paper is, right?

[00:25:22] Speaker 2: Sounds good.

[00:25:23] Speaker 1: It is, uh, it is basically characterized biochemically. Your enzymes, you should have, uh, a few. For some reason, I think I have more notes. If you wanna get more notes, but I, I just stopped at four. We can try more notes, right? Like, this is the present day and ancestral states. And basically, this range is basically showing you, like, how the closer to zero is, the closer to the bottom is, that means that that protein is more, uh it's more, uh it looks more canonically like those charac well-characterized enzymes, right?

[00:26:01] Speaker 2: Mm-hmm.

[00:26:02] Speaker 1: So that means that it is more, uh, more like this can uh, let's for the sake of argument, let's mind this is a car uh, mannase, right? Like, m it chooses mannose, right? So, like, somewhere here, it's, like, kinda looks like a mannose. It really looks like a mannose, right?

[00:26:21] Speaker 2: Mm-hmm.

[00:26:21] Speaker 1: So that's the thing about this, like, like that, right?

[00:26:23] Speaker 2: It's like similarity.

[00:26:24] Speaker 1: So, like, more uh, and this is slightly more complicated than that 'cause it's based on, like it is based on a structure and, like, some language models that.

[00:26:33] Speaker 2: Yeah.

[00:26:34] Speaker 1: I go from company.

[00:26:35] Speaker 2: Prof, quick question.

[00:26:36] Speaker 1: But I think yes.

[00:26:37] Speaker 2: Is, is that graph based on the structure? Is it based on, like, the calculated structure, or is it based on something more like upstream, more like genome-level stuff?

[00:26:48] Speaker 1: Uh, it's, it's basically it's basically a structure.

[00:26:50] Speaker 2: Structure. I see. So more like structural prediction.

[00:26:52] Speaker 1: Yeah.

[00:26:53] Speaker 2: Got it.

[00:26:53] Speaker 1: Yeah.

[00:26:54] Speaker 2: Yeah. That, that makes sense.

[00:26:54] Speaker 1: It's based on proteins. Like, they train the model on a, a miscellaneous number of proteins, right? And that's basically, uh this is this is called, like the value here is actually perplexity. Uh, the perplexity is how sure is your language model of the next token.

[00:27:12] Speaker 2: Mm-hmm. Mm-hmm.

[00:27:13] Speaker 1: Right? So if it's low if it's very sure, it's low perpe low perplexity, right?

[00:27:18] Speaker 2: Mm-hmm.

[00:27:18] Speaker 1: So it's close to the bottom. That means that the language model kinda knows what's gonna be the next sequence. It, it starts with an A, it starts with a, um let me know it. A W, a P, a Q, blah, blah, blah. My language model is fairly certain that the next is gonna be an S, right? And that means that it's, it, it knows this protein family really well. So I know this is a mannase,

right, for example, right?

[00:27:46] Speaker 2: Yep. So.

[00:27:47] Speaker 1: Or, uh, or, uh.

[00:27:49] Speaker 2: So quick question.

[00:27:50] Speaker 1: So the perplexity is good. Like, the protein is, like, is, is guessing well what's the, the protein language is guessing well what's, what's happening.

[00:27:56] Speaker 2: So if we're zooming back, what we have right now is, uh, we are trying to purify protein with, like, HeStack purification, right?

[00:28:04] Speaker 1: Yes.

[00:28:04] Speaker 2: So after purification, we supposed can know more about these enzymes by testing them.

[00:28:10] Speaker 1: Yes. You're gonna characterize them. You're gonna characterize them.

[00:28:12] Speaker 2: Okay. I see. So that's.

[00:28:13] Speaker 1: Right now, this is right now, this is, like, a model.

[00:28:17] Speaker 2: Yeah.

[00:28:17] Speaker 1: The model is, is telling me, like, okay, ancestral states seem to be more, like, broad range, right?

[00:28:25] Speaker 2: I see.

[00:28:27] Speaker 1: And, like, modern, like, extent, like, what's happening right now seems to be more narrow, right?

[00:28:33] Speaker 2: Mm-hmm. Mm-hmm.

[00:28:34] Speaker 1: We're gonna characterize them biochemically.

[00:28:36] Speaker 2: I see. So that's, that's one. That's 1B, right? That's 1B about, like, the phenotype.

[00:28:41] Speaker 1: Yes.

[00:28:41] Speaker 2: So 1C is that it's something that we want to do, but we don't have time to do yet, right?

[00:28:47] Speaker 1: Yes.

[00:28:47] Speaker 2: It's something in the future. So.

[00:28:48] Speaker 1: Yes. I don't think you, you don't have this, but you I think you can have you have time to do some of this.

[00:28:54] Speaker 2: Yes.

[00:28:55] Speaker 1: For your, uh.

[00:28:56] Speaker 2: We can do a 2x3 matrix.

[00:28:57] Speaker 1: For the jamboree.

[00:28:58] Speaker 2: We can do a 2x3 matrix.

[00:29:01] Speaker 1: So, yeah, 2x3 is actually, no. I was thinking it's I think you can have more than that.

[00:29:06] Speaker 2: Yep.

[00:29:07] Speaker 1: You can just have I think it would be really nice. You have

the phage.

[00:29:10] Speaker 2: Yep.

[00:29:11] Speaker 1: You can challenge that high titer phage against all of your isolates.

[00:29:16] Speaker 2: Uh-huh.

[00:29:17] Speaker 1: Some of them are gonna kill some of them. It's not gonna kill some of them.

[00:29:19] Speaker 2: I see.

[00:29:19] Speaker 1: That's fine.

[00:29:20] Speaker 2: So.

[00:29:21] Speaker 1: And I think you once you have your protein purified, I know you at least have one of them, right?

[00:29:27] Speaker 2: Mm-hmm.

[00:29:27] Speaker 1: That you should be really close to purifying, right? You're having you have a really nice band, and you have an a lot of concentration. You purify it, and it's in range. So you have to scale that up. And now that you have that protein, you're gonna repeat the exact same experiment that you put, like, on a plate in your culture. You're gonna put, like, a bit of your phage, right?

[00:29:48] Speaker 2: Mm-hmm.

[00:29:49] Speaker 1: And next to it, on a different plate, you're gonna put a little bit of your protein, right? Both of those things should, like, react, right?

[00:29:57] Speaker 2: Yeah.

[00:29:58] Speaker 1: And then, like, if you show that, like, I think that's a fantastic result.

[00:30:02] Speaker 2: Mm-hmm.

[00:30:02] Speaker 1: I mean, ideally, uh, ideally, we would have, like, all of the ancestral states, right? But I don't know if we're gonna get that.

[00:30:10] Speaker 2: Yeah. So it's like, it's like.

[00:30:11] Speaker 1: For example.

[00:30:12] Speaker 2: It's like two different.

[00:30:12] Speaker 1: For the paper, we need those.

[00:30:13] Speaker 2: Yeah.

[00:30:14] Speaker 1: Yes.

[00:30:14] Speaker 2: But for our agent, it's like two different two different matrix. I feel like challenging phage against bacteria is something we did, and definitely something that is, like, more, like, result promise, right? It's, it's more something, like, we know, like, there are going to be a result. But, like, when we're challenging our, like, purified depolymerase against bacteria, right, this step is something that we haven't done before. And this part.

[00:30:39] Speaker 1: Yes.

[00:30:39] Speaker 2: I know I know we have done this in, in the paper. There's only one concern from me is that so let's say depolymerase, right? How can we see if it reacts with bacteria? Because I know it's not, like, something it's not, like, a lysate, or it's not, like, a it's, it's not, like, something that will completely make a.

[00:31:01] Speaker 1: Yes.

[00:31:01] Speaker 2: Bacteria, like, that vanish, right? It's, it's not like that. It's only it's only breaking down the capsule. So I was thinking about, like, how, how can we how can we do that assay? And then if we're.

[00:31:13] Speaker 1: It is it is yeah. It is called it is it's in the paper that I sent you guys, the.

[00:31:17] Speaker 2: Yeah.

[00:31:17] Speaker 1: Deepo analog. Like, you just literally have to have, like, your plate, your, your, your lawn of bacteria.

[00:31:26] Speaker 2: Uh-huh.

[00:31:27] Speaker 1: Growing them happy. And you literally just add a drop I mean, not a drop, like, X volume, uh.

[00:31:33] Speaker 2: Uh-huh.

[00:31:33] Speaker 1: 5 microliters, 10 microliters, of your purified enzyme.

[00:31:38] Speaker 2: Okay. I see.

[00:31:38] Speaker 1: It will not clear it, but it will definitely look like, uh.

[00:31:41] Speaker 2: Different.

[00:31:42] Speaker 1: It will look different.

[00:31:43] Speaker 2: I see. I see.

[00:31:44] Speaker 1: Like, that means that the, the enzyme chewed away the, uh, the, um, the, um, the enzymes, right?

[00:31:55] Speaker 2: Yeah. So, so, so here, here, here's my point. I feel like we should do these two experiment bilaterally because we are currently at the ste at the stage about purifi purifying protein, right? We're, we're currently, like, trying to.

[00:32:07] Speaker 1: Yes. Yes. I think, like, if you have proteins right now, if you already have proteins, I would start right away resurrecting all of your isolates. So you can test in parallel your phage and your protein, your enzyme.

[00:32:24] Speaker 2: Yep. Uh, I think yeah. I'll, I'll, I'll.

[00:32:29] Speaker 1: Okay.

[00:32:29] Speaker 2: I'll also talk to Rafael about it. I'll also talk to Rafael about it.

[00:32:33] Speaker 1: Yeah.

[00:32:33] Speaker 2: We'll, we'll.

[00:32:34] Speaker 1: Yeah, yeah. No, let's, let's see. I think Raf and Rafael is gonna be away early next week.

[00:32:38] Speaker 2: Mm-hmm.

[00:32:39] Speaker 1: But I'll be here.

[00:32:40] Speaker 2: Yep.

[00:32:41] Speaker 1: So.

[00:32:42] Speaker 2: Sounds good. Sounds good. Okay.

[00:32:44] Speaker 1: Let me see the, uh and then for more figures, uh, you have a why don't you start creating by creating a graphical abstract.

[00:32:56] Speaker 2: Yeah. Uh, we can do that. We can do that.

[00:32:58] Speaker 1: Graphical abstract will look like this, right?

[00:32:59] Speaker 2: Uh-huh.

[00:33:00] Speaker 1: Why don't you come up with a graphic I mean, you this is the problem we're trying to solve. I mean, I mean, the problem let's just step back. Like, let's take a look at the 10,000, uh, feet, uh, feet, uh.

[00:33:11] Speaker 2: Above.

[00:33:11] Speaker 1: Picture, right?

[00:33:12] Speaker 2: Yep.

[00:33:12] Speaker 1: Uh, so it is, uh, we can start with antibiotics are, like, bacteria becoming antibiotic resistant. Let's look back to phage.

[00:33:20] Speaker 2: Yep.

[00:33:21] Speaker 1: That's step one, right?

[00:33:22] Speaker 2: Mm-hmm.

[00:33:23] Speaker 1: Phage ha phages are very specific.

[00:33:25] Speaker 2: Yep.

[00:33:26] Speaker 1: Right?

[00:33:26] Speaker 2: Step two.

[00:33:27] Speaker 1: So finding a phage that can find a bacteria, it is hard.

[00:33:31] Speaker 2: Mm-hmm.

[00:33:33] Speaker 1: What if we can ex artificially extend the host range of this phage?

[00:33:39] Speaker 2: Yep. That's the whole plan.

[00:33:41] Speaker 1: Here, here, we are trying to. That is. And then, like, the evolutionary question is, like again, the pure science question is, uh, all of the models.

[00:33:52] Speaker 2: Mm-hmm.

[00:33:53] Speaker 1: All the mechanistic models on phage-shelf interaction are based just on RBPs and receptor binding proteins.

[00:33:59] Speaker 2: Mm-hmm.

[00:33:59] Speaker 1: That's great.

[00:34:00] Speaker 2: Right.

[00:34:00] Speaker 1: But we know that in certain bacteria, there's other axes of variation, right? One of the axes is, like, the depolymerase, for example. We are hypothesizing that the depolymerase sets an extra host range, a ceiling or, like, alternative, uh, uh a ceiling is the best word I can, can think. Like, a, a host range, right, that could potentially also, um the depolymerase acts as a potential range, I guess, as well, right? So we're gonna and, and then we also have in, in, in the same line, we have a hypothesis that maybe it evolves independently, right? And, and based and by that, we also gonna be we're gonna see if, like, maybe ancestral states are, like, more host more right.

[00:34:51] Speaker 2: Mm-hmm.

[00:34:52] Speaker 1: And the current states are, like, more narrow, right? It could be the other way around. Either, either way is the result is very interesting, right, and publishable, right? And we're gonna relate the ho we're

gonna and that's the reason we are gonna we c and we're and they were interested in that question. Again, like, the question is, like, what if we can extend the host range of bacteria.

[00:35:14] Speaker 2: Yep.

[00:35:14] Speaker 1: Of phage somehow?

[00:35:17] Speaker 2: Yeah.

[00:35:17] Speaker 1: Oh, look. Look at this. All of the models focuses on, like, receptor binding proteins and the receptor.

[00:35:23] Speaker 2: Yeah.

[00:35:24] Speaker 1: But the and then so at least on phage, there is an extra axis that is actually poorly explored. At least it wasn't till, like, the d that paper just came, uh, which is very annoying.

[00:35:33] Speaker 2: Yeah. From Stanford, right?

[00:35:37] Speaker 1: Yeah.

[00:35:37] Speaker 2: All right.

[00:35:37] Speaker 1: Yeah. That was annoying.

[00:35:38] Speaker 2: So, uh.

[00:35:39] Speaker 1: That's, that's, that's how it is, right?

[00:35:40] Speaker 2: Yeah.

[00:35:40] Speaker 1: So they're pretty characterized, but they maybe, uh, uh in any case, we're gonna see where it's gonna we're going for after evolutionary questions, right?

[00:35:49] Speaker 2: Yep.

[00:35:49] Speaker 1: So now we know that this is, uh w so we know at least the host range might be at least partially codified by the depolymerase.

[00:36:00] Speaker 2: Mm-hmm.

[00:36:01] Speaker 1: We're gonna and to, to explore that, we're gonna create a, a tree.

[00:36:07] Speaker 2: Mm-hmm.

[00:36:07] Speaker 1: To find ancestral states, characterize the phenotype, and characterize the biochemically, those enzymes, and link them to the phenotype.

[00:36:17] Speaker 2: Mm-hmm.

[00:36:18] Speaker 1: Of the host, right?

[00:36:19] Speaker 2: Yep.

[00:36:19] Speaker 1: And actually, the, the ultimate experiment would be if I add my polymerase in trans, can I extend the host range, right?

[00:36:28] Speaker 2: Mm-hmm.

[00:36:29] Speaker 1: I don't know if we're gonna get to see to the last one by your, uh, uh, your iGEM.

[00:36:36] Speaker 2: Probably not.

[00:36:37] Speaker 1: It should.

[00:36:37] Speaker 2: Yeah.

[00:36:37] Speaker 1: I think you should be able to have, like to show that the, uh, the enzymes that the enzymes that we were able to, to characterize biochemically recapitulate in somehow the phenotype of the phage that we have.

[00:36:52] Speaker 2: Yep. Uh, Prof, one, one quick question. Going, going back a little bit, why do we choose depolymerase? Because that's a because that's a, like, last-studied axis, right?

[00:37:03] Speaker 1: Yeah. It is, yeah. And that, that, that that's, that's how I would set it. That's how I would set it, right?

[00:37:07] Speaker 2: Okay. Okay. And it's also because it's the first step of, of the infection, right?

[00:37:11] Speaker 1: That's how I would set it as well. I would set it, like, three d three ways. It's the leastest study. It is the first step. And it could potentially be manipulated, right?

[00:37:19] Speaker 2: Yes. Okay.

[00:37:20] Speaker 1: And making fake phage.

[00:37:20] Speaker 2: That, that makes sense. That makes sense.

[00:37:22] Speaker 1: Making fake phage is, is hard, right?

[00:37:24] Speaker 2: Mm-hmm.

[00:37:24] Speaker 1: You could potentially make, like, a small phage, like 5, 5251 or whatever it's called, which is, like, 3,000 basepares, right? You can make that, right, in a plasmid. But if a or phage or many phages are in the tens of thousand, uh, uh, kilobases, right?

[00:37:42] Speaker 2: Mm-hmm.

[00:37:43] Speaker 1: Tens of 50 or is it, like, 60 kilobases? 50? I don't remember.

[00:37:48] Speaker 2: Mm-hmm.

[00:37:48] Speaker 1: Right? So making a ph the novel phage is hard, right?

[00:37:52] Speaker 2: Yep.

[00:37:53] Speaker 1: What if I could but, but that one, I could modify it because it could potentially be added in trans, right?

[00:37:58] Speaker 2: Mm-hmm.

[00:37:59] Speaker 1: Uh, maybe that's, that's a good way. So I would sell it in these three, three ways.

[00:38:02] Speaker 2: Okay.

[00:38:02] Speaker 1: Like, it's the leastest study.

[00:38:04] Speaker 2: That makes sense.

[00:38:05] Speaker 1: It is the, um, uh, uh, it could potentially be, uh, uh, manipulated, right, easily, right? And what was the other word? Sorry.

[00:38:22] Speaker 3: Um, Prof.

[00:38:23] Speaker 1: Yes.

[00:38:24] Speaker 3: Yeah. I have a question for the host range. So, like, like so remember, we are starting this whole story from the botanic garden. And so, like, the then the like, so will the host range be applicable to, to, like, how large should it be? Like, so now we only have, like, based on our grow lab during summer, it's only, like, one phage to, to bacteria. And those two are pretty

similar to each other.

[00:38:47] Speaker 1: Yes.

[00:38:47] Speaker 3: Or, like, yeah, right? And so.

[00:38:49] Speaker 1: Yes.

[00:38:49] Speaker 3: And so.

[00:38:51] Speaker 1: I would test it.

[00:38:53] Speaker 3: And so, like.

[00:38:53] Speaker 1: Yeah. I would test it. I ideally, we should have a matrix as big as this, but we are not gonna have it for any paper anytime soon, right? So that's, like, a problem for, like, five generations of students from now having a having a host range that looks like this, right? We do not.

[00:39:11] Speaker 3: Five generations of students.

[00:39:12] Speaker 1: Right? We do not have this, right? But you have.

[00:39:16] Speaker 3: Right. So, like, yeah. I guess my question would be then, like, if we only have these two bacteria and one phage, will that count as a host range? That's clear.

[00:39:25] Speaker 1: Yes.

[00:39:26] Speaker 2: Yeah. That's still that'll still be a host range.

[00:39:27] Speaker 1: Yeah.

[00:39:28] Speaker 3: Yeah.

[00:39:28] Speaker 1: Because the because what we're trying to do originally

is, like, uh, like, this, this, this is, like, long-term goal, right? Right now, first you need to demonstrate.

[00:39:38] Speaker 3: Yeah.

[00:39:38] Speaker 1: First, you need to demonstrate that the phenotype of the phage can be recapitulated by the biochemistry of the of the enzyme, right? That means that the same in, like, the that's the thing you one of the experiments that you need to do is a lawn of your bacteria, add your phage, high-titrate phage, and let it go. And then at the same time, on another lawn, or maybe the same lawn on a on a different side of the plate, add your polymerase. Like, if the phage is killing that bacteria, your depolymerase, wild type, should be able to clear that. That would be the first great result, right? Showing, like, okay, the biochemistry shows that this host shows this activity, and the phenotype, and that it, it recapitulates very nicely the actual host range of the phage, right? That's, that's the first result. And I think that that would be that would be it. Yes. I agree with you. Like, we don't have more than that, but what we have is, is what we have. And then maybe, maybe.

[00:40:45] Speaker 3: So.

[00:40:45] Speaker 1: Yes, sir.

[00:40:46] Speaker 3: So, like, first thing, like, I want to bring up is, like, for the host range, like, it's really likely that we do have more host range. Like, remember, like, in the I think it's in the end of the summer, and I do the host range test. And so it shows, like, the phage is actually could be more effective at killing some other.

[00:41:06] Speaker 1: Some other bacteria, yeah.

[00:41:07] Speaker 3: Yeah.

[00:41:07] Speaker 1: I think.

[00:41:07] Speaker 3: It's just, like, we didn't find the correct or the most

accurate titrate yet.

[00:41:12] Speaker 1: The most accurate.

[00:41:12] Speaker 3: But also.

[00:41:13] Speaker 1: This is weird.

[00:41:13] Speaker 3: Another yeah. Another concern might be, like, also that bacteria is similar or, like, same to the, like, the bacteria we already have. So, uh and, and so I also I understand it as in a way, like, our, like, final goal would be, like, to extend the host range. And not only by just, like, exploring the wild types, like, we gain from the botanic garden, but also from, like, uh so we, we are, like, tracing back to the to the, like, the ancestral note. And then so, so if the, like, the like, through the evolutionary track, if the, like, the previous notes, the notes of, uh, depolymerase can also show the, uh, binding, lysing, or killing phenotype, and then it can, like, sort of prove our goals. Uh, so, like, now we have E1. Like, so I'm I think I'm curious about, like so we have three notes, right? E1, E40, and E, uh, 281.

[00:42:08] Speaker 1: Yes.

[00:42:08] Speaker 3: And which one is closest to our, uh.

[00:42:11] Speaker 1: I should know this right now. I think the closest one is 1, and the farthest one is, is the larger number. I will double-check, and I will send you that right after this, this call.

[00:42:21] Speaker 3: 'Cause I think that is, like, interesting to think about. Like, if we have the purified protein, like, like, for the first time, it's E1. It's also the closest one to the.

[00:42:30] Speaker 1: Yeah. Yeah. E1 would be the closest one. So wild type, E1, and the phage should have very similar phenotypes, right? And there's another thing that I forgot. Like, you also have, like, resistant mutants, right?

[00:42:45] Speaker 2: Mm-hmm. Yep. We haven't sequenced them yet. We haven't sequenced them.

[00:42:49] Speaker 1: We haven't sequenced them, but we can just grow those and try our ancestral, uh, reconstructed.

[00:42:55] Speaker 2: Oh, that's a that's a that's a really good idea. That's a really good idea.

[00:42:58] Speaker 1: Right?

[00:42:58] Speaker 2: Mm-hmm. That's a really good idea.

[00:42:59] Speaker 1: And I think that would be that would be is they're close enough, but maybe my ancestral states, you know?

[00:43:04] Speaker 2: Mm-hmm.

[00:43:04] Speaker 1: And again, like, the question is just, like again, like, you can always start from the from the top, right? The first ques the first sentence, like, bacteria getting resistant to my antibiotics. Let's try phage. Phage takes forever to isolate. I mean, they're very specific. Maybe we can extend the, the host range.

[00:43:23] Speaker 2: Prof, quick question.

[00:43:24] Speaker 1: Most people look at RBPs, yes. And you got a you got a idea.

[00:43:28] Speaker 2: Yeah. Do you think do you think we should sequence the resistance bacteria?

[00:43:33] Speaker 1: Yes. Yes, yes, yes, yes.

[00:43:36] Speaker 2: Yes. Okay.

[00:43:36] Speaker 1: I think we should sequence that. We should sequence that.

[00:43:38] Speaker 2: I, I, I can I can do a delivering still as well. I just need to grab the.

[00:43:42] Speaker 1: Yeah, yeah.

[00:43:43] Speaker 3: Do it. Do it.

[00:43:44] Speaker 1: We have we have the pellets. We have the pellets. I don't think we did the extraction. Carol, you did the pellets a long time ago.

[00:43:51] Speaker 3: Yeah, but where are the like, are they in like, no.

[00:43:55] Speaker 1: They're minus 20. They should be in minus 20, yes.

[00:43:58] Speaker 3: If that's in minus 20, then it's really likely that it's not in minus 20. 'Cause I didn't see them in minus 20. Will, will there be any possibility that we move it to minus 80?

[00:44:10] Speaker 1: I doubt that, but you can you can check. You can check.

[00:44:12] Speaker 3: 'Cause, 'cause Rafael once cleaned the minus 20. And I think now.

[00:44:18] Speaker 1: He does like to clean. He does like to clean. So label everything 'cause he really likes to clean. Yeah.

[00:44:24] Speaker 3: But, like, the, the point is, last time, when I opened the minus 20, I don't think I.

[00:44:30] Speaker 1: Yeah.

[00:44:30] Speaker 3: I saw any.

[00:44:31] Speaker 1: But, but the and I'm trying to remember if you did the extractions. If you did the extractions.

[00:44:36] Speaker 3: It's time to check our summer notesbook.

[00:44:39] Speaker 2: Yeah. I, I think Carol probably did the extraction because I remember it was it was a day my, like, flight got canceled by a typhoon.

[00:44:45] Speaker 1: Yeah. Like, if you did the extractions, the, the DNA might be in, in the 4-degree, which is fine. It, it stays for for example, uh, Alex's, uh, plasmids were dropped Friday, but were not picked up until Tuesday. So they were on in that box, getting one for four days.

[00:45:08] Speaker 2: It was still fine. It was still fine.

[00:45:10] Speaker 1: Yeah.

[00:45:10] Speaker 2: It was still fine.

[00:45:11] Speaker 1: So this so it's gonna be fine. Which reminds me, if Alex drops them, uh, maybe try to Caltech.

[00:45:18] Speaker 2: Yeah. Maybe Caltech is, like, a faster, faster place to.

[00:45:23] Speaker 1: Also, since I'm not gonna be there, Alex, could you buy bleach, and I'll pay you back?

[00:45:28] Speaker 2: Okay. I'll, I'll buy bleach. The, the like, the bottle from Target, right? Okay. I'll, I'll figure it out.

[00:45:33] Speaker 1: Yeah, yeah. Just, just yeah, yeah.

[00:45:34] Speaker 2: Okay. I'll figure it out.

[00:45:36] Speaker 1: Thank you. No, 'cause you 'cause you have a car, right?

[00:45:38] Speaker 2: Yeah, yeah. I do have a car.

[00:45:39] Speaker 1: I'll pay you back. I'll pay you back. I'll pay you back.

[00:45:41] Speaker 2: Of course. No, no worry about it.

[00:45:43] Speaker 3: Uh, so sh maybe shall we make a, like, a to-do list before what.

[00:45:48] Speaker 1: Yes.

[00:45:48] Speaker 3: At least what we can do before October, let's say, 20? 'Cause I, I don't think we can get so much things done before 15. Like, which as of which is your expected date to finish?

[00:45:59] Speaker 1: Well, October 20. October 20 is just, like it's just that set? Or, like, would you would it's, like.

[00:46:04] Speaker 2: Yeah. It's that set. That's it. And then.

[00:46:06] Speaker 1: That's it.

[00:46:06] Speaker 2: That's it. Like, like, like.

[00:46:07] Speaker 3: No, but that's the end for iGEM, but not for the project, for sure.

[00:46:10] Speaker 1: Oh, yeah, yeah, yeah, yeah, yeah.

[00:46:11] Speaker 3: Like, what we can do before October. Like, uh, I can text really quick. Like, so, so 21 like, 21st is Wednesday. And so, like and our fabric ends at 20, which means, like, we sort of have more time from.

[00:46:26] Speaker 1: In the area?

[00:46:28] Speaker 3: Uh, October 16 to 20.

[00:46:30] Speaker 1: Yeah.

[00:46:30] Speaker 3: Yeah. So, like and there's, like.

[00:46:34] Speaker 1: Yeah.

[00:46:35] Speaker 3: Two no. Wow. It's actually only, like so this week, next week, and then four days left.

[00:46:43] Speaker 1: Yeah. Yeah.

[00:46:43] Speaker 3: So, like, it is, uh.

[00:46:44] Speaker 1: You.

[00:46:44] Speaker 3: Yeah. So it is, like, half two weeks.

[00:46:47] Speaker 1: You for sure, for sure have you have the ability to create the E1 or the E40, whatever it was, right?

[00:46:55] Speaker 3: E1. E1.

[00:46:56] Speaker 2: E1. It was E1.

[00:46:57] Speaker 1: It's kind of up a little bit. And honestly, it is one of those things, like, if you were able to do this one, you should be able to all to do all of the ancestrals because they are identical, very similar to each other, right? So there is that. And, and I know, I'm it is it is annoying, but, like, when you do this kind of work, it's like, like, unfortunately, like and I was telling Alex, you know, like, when you do a test, a written test, you could do you could five you could fire five questions and still get a name, right? But if you in a lab, if, like, if you if you messed up or you accidentally did something wrong in one step, the whole protocol might, like, might just right? So there is one of those things. So whatever you did for E41, just repeat that same again, right? I don't know.

Maybe it is, uh who knows what it was, you know? It might.

[00:47:45] Speaker 3: So what you said about we can do E1, it's just may, like, we drop, like, one drop of E1 protein onto our bacteria.

[00:47:55] Speaker 1: Yes. Yes. So, unfortunately, you do need a lot of protein. I, I was looking at the paper, and it doesn't say how much. Actually, that would be a really good homework for you guys. Go on Google Scholar or, like, anything and see if you can find how much protein you need to do a spot test. That's what this experiment is called. Spot test.

[00:48:17] Speaker 3: Spot test?

[00:48:18] Speaker 1: Spot SPOT test. Yeah.

[00:48:20] Speaker 3: Uh, we.

[00:48:24] Speaker 1: Spot test.

[00:48:25] Speaker 3: Spot test.

[00:48:26] Speaker 1: Spot test.

[00:48:26] Speaker 3: Spot test.

[00:48:26] Speaker 1: Yeah, yeah, yeah.

[00:48:27] Speaker 3: Like, I think from now on, every time we, we do the, like, before purification, we will have two mil uh, yeah, two mil of, of protein.

[00:48:37] Speaker 1: Yes. Yeah. No, but I think, like, for instance 'cause you've been doing only, like, 50, 50.

[00:48:43] Speaker 2: 50 mil of culture. Yeah. 50 mil of culture.

[00:48:46] Speaker 1: 50 mil.

[00:48:46] Speaker 2: We're doing it.

[00:48:47] Speaker 1: Actually, you're doing 30 mil. But you're extracting from 30 mils. Like, that's what Rafael told me.

[00:48:51] Speaker 2: Yeah.

[00:48:51] Speaker 1: You extract for from 15 Falcon cubes twice.

[00:48:55] Speaker 2: Yeah. So 30, 30 culture in total.

[00:48:57] Speaker 1: Yeah.

[00:48:57] Speaker 2: Only. Yeah. That's correct.

[00:48:58] Speaker 1: So you should be able to extract from, like and that the forms that we have should hold 60 milliliters of culture if all you get the same. It's fine. We can get more columns, or we can get bigger columns. The columns get here in no time. So it's just to say, like, ideally, we should run this experiment from half a liter of, like, culture.

[00:49:24] Speaker 2: Increase the volume.

[00:49:25] Speaker 1: It's not more. It's 10 times.

[00:49:26] Speaker 2: Increase volume. Okay.

[00:49:27] Speaker 3: So is that, like, before like, like, only after extraction where, like, the protein solution also need to be, like, purified, like?

[00:49:37] Speaker 1: Before the extraction? No, it's just after the extraction. Yeah. When you extract.

[00:49:40] Speaker 3: Oh, okay.

[00:49:41] Speaker 1: You, you, you collect it in a column, right? In principle, only the only the protein of interest stays there, and everything else, uh, goes through.

[00:49:48] Speaker 2: So, Prof, what you mean is that when we when we grow and induce a culture when we induce a culture, we induce with a volume of 500 mil of culture, right? Is that correct?

[00:49:59] Speaker 1: Yes. Yeah. So far, you've been doing 50 mils.

[00:50:02] Speaker 2: Yeah.

[00:50:04] Speaker 1: You need, like, half a liter.

[00:50:07] Speaker 2: Okay.

[00:50:08] Speaker 1: Yes.

[00:50:09] Speaker 3: Do we do we have the incubator to do some?

[00:50:11] Speaker 2: Yeah. I don't think we have.

[00:50:13] Speaker 1: But you can have worst-case scenario, you can have, like, uh, multiple flasks, right?

[00:50:20] Speaker 2: Yeah. That, that.

[00:50:21] Speaker 3: Okay.

[00:50:21] Speaker 2: That, that works. Yeah.

[00:50:24] Speaker 1: Yeah. You can have multiple flasks. I mean, it's just, uh but, like, if you have the protein, why don't you why don't you I mean, that's the kind of, like, I'm not sure. So that would be a really good homework for all of us to find how much.

[00:50:37] Speaker 2: Protein we need.

[00:50:38] Speaker 1: At any time you need for a.

[00:50:41] Speaker 2: Spot test.

[00:50:41] Speaker 1: A spot test.

[00:50:42] Speaker 2: Yeah. Sounds good.

[00:50:42] Speaker 1: Right? And you know how much protein you got from, uh, 50 mils. 15 from 15 mils. Sorry.

[00:50:51] Speaker 2: Yeah. We know how much we got from 50.

[00:50:54] Speaker 1: Let's see. Let's see. Uh, okay. Hold on. What's happening? I can't open my WhatsApp. Hold on. Let me see it in my phone. Let me one give me one second.

[00:51:16] Speaker 2: Yeah. It's fine.

[00:51:18] Speaker 1: I need to update WhatsApp. That's the reason it's not doing.

[00:51:22] Speaker 2: Should we call it a day after this? Should we call it a day after this part?

[00:51:29] Speaker 1: Yeah, yeah. No. I mean, like I mean, it is just, like why don't you, uh like, you uh, why don't you come up why don't you send me your graphical abstract just to make sure we're on the we're on the same page, right? Right away, uh, and then you can use this, the phylogenetic tree. You can use a cartoon if you want. We can replace them later for the real thing.

[00:51:51] Speaker 2: Mm-hmm.

[00:51:52] Speaker 1: Uh, um, uh, yeah. And then find right away how much

protein you need to do your, uh, your protein assays, right? And start planning. I mean, you might already have enough protein to start, like, a small test, right? Hopefully, you realize that you still need to you need to revive all of the isolates that you had at some point, right?

[00:52:17] Speaker 2: Mm-hmm.

[00:52:18] Speaker 1: So you know what to do, right? You, you revive them and say, "Fine, well, you were isolating phages. Uh, just, just keep them, uh, growing. Nothing else has to happen to them. They just have to grow, right? And start making, uh you just have to make a loan of them and add your hi-hi tighter phage, just a drop, and it should clear. And then, uh, if you have enough protein already, why don't you go ahead and do a test?

[00:52:48] Speaker 2: Sounds good.

[00:52:49] Speaker 1: You should be able to do a test right now with the E1, for example, right?

[00:52:53] Speaker 2: Yeah. Uh, I think what would be best is that I will, uh like, we will go talk to Rafael about timeline, and we can figure out, like, what day we can do which because we only have, like, some limited amount of days.

[00:53:04] Speaker 1: Yes. Yeah. I'll, I'll message him. I'll message him. He's very he's apparently, we haven't gotten kanamycin, which is annoying.

[00:53:12] Speaker 2: Yeah. Can we borrow more?

[00:53:13] Speaker 1: One of the things you, you can borrow kanamycin, but at the same time, you do I one of the things, like, I'm gonna I'm gonna insist. You don't need kanamycin anymore. You don't need kanamycin to select your transforms.

[00:53:26] Speaker 2: Okay. Yeah.

[00:53:27] Speaker 1: Once you know there's nothing else there, and if you're

doing everything in aseptic technique perfectly well, nothing else should be in your culture, right?

[00:53:36] Speaker 2: Mm-hmm.

[00:53:36] Speaker 1: 'Cause you the kanamycin was just to select your transforms the first time.

[00:53:40] Speaker 2: Yep.

[00:53:41] Speaker 1: Right?

[00:53:41] Speaker 2: Mm-hmm.

[00:53:42] Speaker 1: If I mean, 'cause you're gonna trans I and he's not transforming anymore 'cause you have already transformed and stuff.

[00:53:47] Speaker 2: Oh, the ones. Yeah. It should be fine.

[00:53:51] Speaker 1: Yeah. The ones that just the, the second set. Not the first set. The second set.

[00:53:54] Speaker 2: The second set.

[00:53:54] Speaker 1: It is. You have transformed stuff. You, you can grow those right away, and they, like and then were you were you you were you able to do the are you familiar with the transform with the induction protocol?

[00:54:09] Speaker 2: Yeah.

[00:54:10] Speaker 1: Like, could you do it on your own?

[00:54:11] Speaker 2: I think so.

[00:54:12] Speaker 1: Could you do it on your own?

[00:54:13] Speaker 2: Yeah. Yeah. We're, we're pretty sure about induction.

[00:54:14] Speaker 1: There's, there's no need to Rafael then, right? I mean, he can help if he needs if you guys need it. But you literally just all you need is, like, you, you know your transform stuff. You don't need kanamycin anymore. Well, you might need for the for the first plate, but you in principle, you don't need them anymore. Maybe, maybe if you have plates with kanamycin, you can extract them on them. You know whatever grows there is your transformed plate. Start growing. You're starting culture. You're you pick culture, induce, extract, run a gel for sake of, uh, mental oh, I think I did it right. If you're in a rush, just, like.

[00:54:55] Speaker 2: Just.

[00:54:56] Speaker 1: Extract with.

[00:54:57] Speaker 2: Just extract.

[00:54:58] Speaker 1: Just go to purification. Just go to purification.

[00:55:00] Speaker 2: Sounds good.

[00:55:00] Speaker 1: And after purification, you can run a gel after. I mean, that would be nice just, just to make sure.

[00:55:06] Speaker 2: Sounds good.

[00:55:06] Speaker 1: You have something there 'cause what you think that you have protein, and then you add it to your cultures, and nothing is being, uh, uh nothing is happening to the cultures, right?

[00:55:16] Speaker 2: Sounds good. Thank you so much, Prof.

[00:55:19] Speaker 1: Please. Yeah. No. Yeah. And then, like, you can coordinate them on yourself. If you know how to do the, the, the, the induction, by all means.

[00:55:26] Speaker 2: Okay.

[00:55:26] Speaker 1: I'll, I'll message Rafael as well so, so he knows what's up.

[00:55:30] Speaker 2: Yep.

[00:55:31] Speaker 1: And then and then just send me right away the your graphical abstract to so we're on the same page.

[00:55:38] Speaker 3: What abstract?

[00:55:39] Speaker 2: Graphical. So it's like a flowchart.

[00:55:43] Speaker 1: Like the like, the, uh, a graph like, the question. Bacteria are annoying. Um, phage.

[00:55:53] Speaker 3: What do you mean? Like, do you want us to, to send, send you, uh, like, a graph of our thought on this product?

[00:55:59] Speaker 1: Yeah, yeah. So the for iGEM for iGEM, you do need to submit a graphical abstract.

[00:56:04] Speaker 3: Oh, okay.

[00:56:05] Speaker 1: Yeah. So I would like to see a draft of that just to make sure we're on the same page, just to make sure that we that I communicated, like, what idea the idea of what I think the project is, right?

[00:56:16] Speaker 2: Yeah. Sounds good.

[00:56:17] Speaker 1: And, and I remember to sell this. I was like, "Yeah. This is a proof of principle, right? We're trying to engineer ways to extend the whole strange of phages. Uh, the polymerases are not explored widely. Are easy to address? " And here we are.

[00:56:36] Speaker 2: Sounds good. Thank you so much, Prof. Yeah. Go ahead, Carlos.

[00:56:39] Speaker 3: Another thing that so if we are going to sequence the, uh, resistant bacteria, like, if we get the genome, like, by next week, and are we going to include that for our iGEM deliverable? And if so, like, what kind of, like, conclusion can we make based on that? I, I believe that will be some.

[00:57:00] Speaker 1: Yes. And that's an excellent question. So that's, uh and I'm glad you, you, you brought this up. You should be able to, from the genome, to have a really good guess what carbohydrates are in the surface. Like.

[00:57:14] Speaker 2: Sure.

[00:57:14] Speaker 1: Let's do some of that.

[00:57:16] Speaker 3: Is that software-wise?

[00:57:17] Speaker 1: Yes. Do some homework on that. If no, I'll, I'll, I'll, I'll give you guys. Basically, you, you could potentially one of, one of the things I can think you could do is, like, a circle spot highlighting the cassettes, right, that each of your, uh, your genomes might have, right? Oh, this one produces, uh, this particular sugar. This pro this particular sugar because there's the cassette, the biosynthetic cassette to make that sugar, right?

[00:57:44] Speaker 2: Yeah.

[00:57:45] Speaker 1: So you should be able to predict those. So and, and that could be an extra link, right? Yes.

[00:57:52] Speaker 3: So I guess, like, before October 21st, what, what we, like, are gonna do is we ha we will do the E1, like, the three nodes, do the spot test plus the CDS12 wild, wild type, like, the wild type of the column, which is CDS12, right? And this is the first thing. And second is, like, uh, we, we want to

get the resistant bacteria sequence back ASAP, and then and then do the genome stuff, which means with better with better check where are the pallets and if it's being extracted tomorrow. And then.

[00:58:29] Speaker 1: Yes. I, I will check that right away. I will check that right away. I think you did extract that. I think you did.

[00:58:35] Speaker 3: Yeah. I also 'cause it sounds familiar to me. Like, I, I think I extract so many things, but.

[00:58:41] Speaker 1: No. I think you did extract it. So if you did extract it, it's definitely in the minus 20, in the fridge part, that is.

[00:58:48] Speaker 3: Then.

[00:58:49] Speaker 1: No, no, not in the minus 20. In the 4. In the 4. Sorry, sorry, sorry. In the 4.

[00:58:53] Speaker 3: 4 or.

[00:58:53] Speaker 1: In the 3.

[00:58:55] Speaker 3: Oh, in 4. Um, I don't know if, if Alex or Oliver could, could quick check 'cause, 'cause if so, it will be in the rack with a lot more.

[00:59:05] Speaker 1: Yeah. It will be in one of these racks. Yeah. It will be in one of those racks. Yes.

[00:59:09] Speaker 2: I was in I was in the restroom. Sorry about that.

[00:59:11] Speaker 3: Oh, you're late. You just. You are what?

[00:59:15] Speaker 2: I was in the restroom.

[00:59:18] Speaker 3: Oh, oh, oh. Yeah. We just said, like, if you or Olivia are available, could you guys please do a quick check at our 4 degrees.

[00:59:27] Speaker 2: Yeah.

[00:59:27] Speaker 3: To see to the fridge to see if the pallets gosh, but I, I remember I, I forgot the, the, the, the number for the resistant ones. But I think it's in the.

[00:59:37] Speaker 1: Maybe it's, it's I, I hope it's in the number. Yes. Yes.

[00:59:41] Speaker 2: Yeah.

[00:59:41] Speaker 1: It should be in the number. Yes.

[00:59:43] Speaker 3: It's definitely in the, uh, our group chat. Yeah.

[00:59:47] Speaker 2: Yeah.

[00:59:48] Speaker 1: Oh, and the group chat as well. Yes.

[00:59:51] Speaker 2: Sounds good.

[00:59:51] Speaker 3: So.

[00:59:52] Speaker 2: Okay. I will check tomorrow morning. That's the that's the first thing I will do.

[00:59:56] Speaker 1: Yeah. And then, like, in like, hopefully, you realize, like, in parallel, you can start reviving the cells.

[01:00:01] Speaker 2: Mm-hmm.

[01:00:02] Speaker 3: Yeah.

[01:00:03] Speaker 1: 'Cause you need you need to have.

[01:00:05] Speaker 2: All strange.

[01:00:06] Speaker 1: Whatever number of bacteria you isolates, you have as loans, right?

[01:00:09] Speaker 3: Wait. But Rafael threw away all the, um, LB sure LB plates.

[01:00:14] Speaker 1: What?

[01:00:16] Speaker 3: He said, like, even though we parafilmed it in the fridge, but it's been a whole summer, and we put a lot of stuff into it. And, and also, we didn't have many in that, I think. Yeah. But, but he do, like, throw.

[01:00:30] Speaker 1: He doesn't that is that is unfortunate. So, so there it is. You, uh, start right away. You need to make media and plates to start reviving them. I know.

[01:00:41] Speaker 2: All right. Lots of work. Uh.

[01:00:44] Speaker 1: Bring, bring more people, guys. Bring more people, right?

[01:00:47] Speaker 3: Shall we like, what about Vincent and CC? Uh, not CC. Vivi.

[01:00:54] Speaker 1: I, I my understanding is, like my understanding is, like, there was no, like, uh, match with you guys with, uh, Vincent, right? Which is fine. I mean, so he's not.

[01:01:05] Speaker 2: I mean, like, if you can if you can do plates and help with us with, like, lab works, like, yeah.

[01:01:11] Speaker 1: Oh, there it is. So th-th-then let's, let's, let's make it let's make it happen. He needs to learn from you guys. So, so th-there is that, right? So it's gonna take a couple of days.

[01:01:19] Speaker 3: Yeah. I've teached, uh, what's her name? Maggie every summer.

[01:01:24] Speaker 2: Yeah.

[01:01:24] Speaker 3: But I need to re-refresh my mind about how to do those stuff.

[01:01:29] Speaker 2: Yeah. Sounds good.

[01:01:30] Speaker 1: Yeah. So th-th-that's an option. And, and then the because I agree with you. I agree with you. This is, like, uh, making me plates and medias is gonna take some time. You can do that. You can do that t-tomorrow, right? And you have enough for, like, to get you started. And then you start you start striking the bacteria, start growing them, right? You should have, like, 27 cul maybe not 27 cultures at the same time, but, like, a good number of cultures going on at the same time. So you, you can do your spot assays with your, uh, the phage and the D-polymerase, right? Ideally, ideally, each plate is gonna have five circles, right? The phage, uh, and all of the nodes, right? To see, like, oh, maybe, uh, the, the D-polymerase wild type behaves exactly like the phage, uh, and then others do not as much, but maybe have bigger whole strange, right?

[01:02:28] Speaker 3: Oh, let's see what will happen after we do the spot test. Hopefully, it will give us some positive feedback.

[01:02:34] Speaker 2: Hopefully.

[01:02:35] Speaker 1: Yeah. No. I think the spot test should look it should look the same, like, the what the phage should look the same as the wild type, for sure, right? If, if the others one look different, it's still a result, right? We but for sure, the wild type and the phage should look the same. And that should be one of your main figures, for example, right? And speaking of figures, to include, I mean, you have some there's at least one gel that's very nice. I will use that gel for your.

[01:03:08] Speaker 3: Yeah.

[01:03:09] Speaker 1: All right.

[01:03:09] Speaker 3: So I think, yeah, we've been, like, spending maybe a little bit too much time, um, trying to prove we have.

[01:03:16] Speaker 1: Tweaking, tweaking the polymer. Tweaking, tweaking that. But you got one right. The E1 is perfect.

[01:03:21] Speaker 3: Yeah.

[01:03:21] Speaker 1: Just like just what you did with that one. Fry with maybe, maybe the problem is you're trying too many at the same time, right? If you do just one at a time, I mean, 'cause you did all of them, right?

[01:03:40] Speaker 2: We'll see. We'll see. We'll see. We'll see.

[01:03:42] Speaker 1: You guys can you guys can do it. You guys can do it, right? I mean, I need my paper, and you need your iGEM, so.

[01:03:51] Speaker 2: That's all right. That's all right. We got it. We got it.

[01:03:53] Speaker 3: So shall we coordinate with Vivi at.

[01:03:58] Speaker 2: Uh.

[01:03:58] Speaker 3: Yeah. That's perfect.

[01:03:59] Speaker 2: I will, I will.

[01:04:00] Speaker 1: Like, if you don't mind teaching them, I will go for it.

[01:04:03] Speaker 2: Okay.

[01:04:03] Speaker 1: I will go for it. 'Cause in the case, you have all these

people doing there, you know.

[01:04:06] Speaker 2: Yeah. We need some backup right now.

[01:04:07] Speaker 1: Even if they are just even if any of them is just there, like, making media and plating, that's a lot of help, right?

[01:04:15] Speaker 2: Yep.

[01:04:15] Speaker 1: And maybe, may-maybe that's, that's the easiest stuff. They do the easiest stuff. And maybe you do the advanced stuff, which is the biochemistry, right?

[01:04:22] Speaker 2: Yeah. Spot test and then protein purification stuff.

[01:04:26] Speaker 1: Yeah.

[01:04:26] Speaker 2: Yep.

[01:04:28] Speaker 3: Oh, wait. I have a last-minute question. So, so you said, like, when you trace back on the evolutionary evolutionary track and to see what if, like, uh, will the ancestral nodes go, uh, like, show the similar phenotype to the to the wild type? And so if the phenotype test does show the similar result, then, like, like, what kind of, like, conclusion we want to make from this evolutionary track?

[01:04:55] Speaker 1: Yeah. And our hypothesis is, like, ancestral stage was broader.

[01:05:00] Speaker 2: It's broader.

[01:05:00] Speaker 1: And, um, Vincent and Vincent is just, like, is very narrow, right? So in principle.

[01:05:06] Speaker 3: But yeah, I.

[01:05:07] Speaker 1: That's the hypothesis.

[01:05:08] Speaker 3: I think.

[01:05:09] Speaker 1: Might be wrong, right?

[01:05:10] Speaker 3: No. No. Uh, like, but, but, like, I think we are gonna to explain, like, why this happened. Like, we are going to attribute that to some, like I don't know if that's, uh, like, on the sequence side or, like, whatever biochemical properties of the ancestral nodes. I think my question will be, like, it-it's more, like, like, are we going to analyze those difference or, like.

[01:05:33] Speaker 1: Hey, for, for the paper, yes, right? For this one, it's just, like, yes, you were saying, like.

[01:05:38] Speaker 3: Oh.

[01:05:39] Speaker 1: Ideally, maybe ideally, maybe we go back to, like, looking at the you can generate the structure of the ancestral states, right? And maybe particular residues are, like, the ones that are, like, extending the range, right?

[01:05:53] Speaker 3: Okay.

[01:05:53] Speaker 1: Like, you put like, if you make a complex with a, a compilement of the, the, the enzyme with a particular sugar, right? Maybe different sugars have different yes. I agree. Like, we should try to be, like, uh, explore the causality of it, right? But I think, for now, I honestly, like, if you have if you have the phage and also, it's about learning, right? Maybe you learn that, uh you learn some stuff, I hope, right?

[01:06:21] Speaker 3: Yeah. Yeah.

[01:06:22] Speaker 2: So.

[01:06:22] Speaker 1: For sure.

[01:06:23] Speaker 3: For sure.

[01:06:23] Speaker 1: So reflect, reflect, reflect on that on your w in your wiki, right?

[01:06:28] Speaker 2: Yeah.

[01:06:29] Speaker 1: Right. Yeah.

[01:06:31] Speaker 2: Okay.

[01:06:33] Speaker 1: You reflect on that. And then, like, uh, let me, um I'll put together some methods for the computational part, right, which is, like, the tree, uh, and the LLNs.

[01:06:46] Speaker 3: And also the, like, the genome assembly. Like, I, I, I don't know if we cover that in the chat, but, like, but, like.

[01:06:52] Speaker 1: Oh, that's true. That is true.

[01:06:53] Speaker 3: The, the assemble from metagenome and how we analyze, how we get the, uh like, how we come up with the CDS 10, 11, and 12.

[01:07:04] Speaker 1: Yes. That's true. We did BLAS. It was just BLAS. Oh, that's a that's great, actually. That's a good question. It's BLAS. And I think you should discuss that at length. That's a that's a really good observation. I think that's a really good, like we, we did we, we assembled the genome from Illumina. It's fully closed. We call proteins, right? And then, uh, we by BLAS, we assign them some taxonomy, right? But by comparative genomics and there's the paper that I sent you guys on the chat. That's where this figure comes from. Hopefully, this figure looks familiar. It's in one of the papers. It is the, uh, Frontiers. By comparative genomics, we think it we think we don't have TEM, unfortunately. We should have TEM for the paper, right? But we don't have TEMs yet. So by comparative genomics, we think it looks like this.

[01:07:55] Speaker 2: And we also and we also, like, use RBP detect to figure

out which protein we're looking at. And CDS 10, 11, and 12 was, like, on the top of the list, but they're actually not receptor-binding protein. They're actually D-polymerase. That's also how we come to those three.

[01:08:11] Speaker 1: Yeah. And then, on top of that, I would definitely, uh, add, uh, the, the prediction 3D folds like this, right?

[01:08:19] Speaker 2: Yep.

[01:08:21] Speaker 1: And follow this cartoon to, uh, to, to illustrate yourself. Again, this will be the CDS, uh, not the, the 11, which is the actual receptor-binding protein. It will be this one. And then, by comparative genomics, that is looking at other T4 phage T this is a polyphage. So let's say that it is a you remember, at some point, you have to say, like, we don't have TEM, unfortunately. But by comparative genomics, we posit that it's a polyphage, right? That it looks like this. The polyphage and the T we then we have myos with the encoder viralis. We have myoviruses, which is the T that's the one in your in your iGEM sweatshirt in the back. That's a T4, and that's, uh, myo. A, a, a cipho is, like, almost like a myo, but with a, uh, curved tail. And the podo is tiny like this. It's just, like, the circle, which is little, little, uh, legs.

[01:09:23] Speaker 2: Mm-hmm.

[01:09:23] Speaker 1: So this will be a so this is by comparative genomics, we think it, it, it is.

[01:09:27] Speaker 2: Looks like that.

[01:09:27] Speaker 1: You know, it's a, a polyphage, right?

[01:09:29] Speaker 2: Yeah.

[01:09:29] Speaker 1: And then you can, uh on, on that node, you can literally make, uh, a figure that looks like this.

[01:09:35] Speaker 2: Mm-hmm.

[01:09:36] Speaker 1: Right? Find other polyphages that look like that, similar polyphages, right, that you can say, "Okay. Yeah. Yeah." So the so your reviewer is convinced, like, we don't have TEM, but we have there are some phages that are similar enough out there that we know that our phage looks like this. But yeah, that's, that's a good observation. We should mention, uh, how we came up to, to this, right? And then again, like, just a, uh just this, like, taxonomy with, like, BLAS, right? A-annotation with BLAS highlighted this, right? And there's one more thing that I ha we haven't discussed, but let's, let's discuss as, as we come this later. It's, like, many of the D-polymerases are normally attached to receptor-binding protein. And but ours is attached it's, it's on CDS. So that could potentially be interesting because it has the different, uh the whole chain.

[01:10:35] Speaker 2: Structure.

[01:10:35] Speaker 1: Could potentially. Right? But it's not a gene transfer. As opposed to if the D-polymerase was part of the CDS, maybe it's the evolution is more restricted. So something like, uh, uh, this mentioned in the paper or something, you know? But, like, for this one, it's just I don't know how, how important that is. But yes. I agree.

[01:10:55] Speaker 2: Sounds good.

[01:10:55] Speaker 1: I think you have enough to you have enough for your wiki. You have enough to write, like, the phages, where they come the phages, uh, uh, where the, the sampling, right?

[01:11:04] Speaker 2: Mm-hmm.

[01:11:05] Speaker 1: We got the samples from here. We have n number of genomes, right? Describe the genomes. Include your, uh, uh, um, circle spots, right?

[01:11:16] Speaker 2: Mm-hmm.

[01:11:16] Speaker 1: Many of those things you can do online. So just, just, just, just find the tools for that. Uh, again, the ph the figure, uh, the, uh, isolation, the whole strange that we have, again, the comparative genomics of the phage, you know? All of that should already be in the wiki, right? Then sorry. I'm sorry, guys. We were in crunch time, so.

[01:11:40] Speaker 2: Yeah.

[01:11:40] Speaker 1: As you know. As you I mean, such is life. Such is life. Such is science.

[01:11:46] Speaker 3: Oh, and also for wiki, I feel like I, I saw Olivia, uh, like, the website has already have so many subsections. So, like, if it's possible, maybe we can, like, assign.

[01:11:57] Speaker 1: Let's do it.

[01:11:58] Speaker 3: Some parts to different.

[01:11:59] Speaker 1: Yeah.

[01:12:00] Speaker 3: Yeah.

[01:12:00] Speaker 1: And I'm happy.

[01:12:01] Speaker 3: That's fine.

[01:12:01] Speaker 1: Happy to write on the wiki as well. So.

[01:12:03] Speaker 2: Mm-hmm.

[01:12:04] Speaker 1: Let me know. I'm happy to, to, to start writing down.

[01:12:06] Speaker 2: I think, like.

[01:12:07] Speaker 3: I think maybe it's better like, it's better to first send out

what sections do we need to fulfill and, and then assign to different people.

[01:12:15] Speaker 1: Yes.

[01:12:15] Speaker 3: Which will make things more efficient.

[01:12:18] Speaker 1: Yes.

[01:12:18] Speaker 2: So that's.

[01:12:18] Speaker 1: Great.

[01:12:19] Speaker 2: That's great.

[01:12:19] Speaker 1: Great.

[01:12:19] Speaker 3: Just, just something to bring up for this, uh, like, what we should now appreciate.

[01:12:25] Speaker 2: Yeah. I think, like.

[01:12:25] Speaker 1: Yeah.

[01:12:25] Speaker 2: About, about writing wiki. I think Prof can definitely take care of the software section first because Prof is definitely the person who, like, knows how it works and.

[01:12:34] Speaker 1: I agree. I agree that part. And now, I'll, I'll, I'll you can look at the code. It's very easy to follow. It's just, like, I'm familiar with, like, the tools, right?

[01:12:41] Speaker 2: Yeah. I think, like, I'll, I'll, I'll we'll, we'll talk about, like, how can we divide and conquer the wiki. Yeah. For sure.

[01:12:50] Speaker 1: Yeah.

[01:12:50] Speaker 2: Agree.

[01:12:50] Speaker 1: So I think, like, I will send that ASAP, right? Like, if you have, like, time today, uh yes. I'm so I'm so sorry, guys.

[01:13:02] Speaker 2: Why?

[01:13:02] Speaker 1: I know you guys are, like but it's such is such is the student life.

[01:13:06] Speaker 2: Yeah.

[01:13:06] Speaker 1: Hopefully, you enjoy the student life. And this is fun, right? You learn stuff. You do stuff.

[01:13:11] Speaker 2: For sure.

[01:13:12] Speaker 1: Stuff like that. And also, like, it helps you, like, if you can meet, uh once when, when you meet with the president, you know, maybe you can show the wiki already, tell me.

[01:13:21] Speaker 2: Yeah.

[01:13:21] Speaker 1: You know, I'll g.

[01:13:22] Speaker 2: Prof, I will send you a time later and see if you can come over to the meeting as well.

[01:13:26] Speaker 1: Oh, yeah. No. I would love to. I would love to. Yes. I would love to. And then the and the paper, right? Say, like, "Oh, look at this. We already have a paper. "

[01:13:32] Speaker 2: Uh-huh. Yeah.

[01:13:34] Speaker 1: Right? And we have at least two more coming up. Maybe the phage genome, right, and the actual biology, right?

[01:13:43] Speaker 2: Yeah.

[01:13:43] Speaker 1: That will be two. Or maybe the biology with the phage genome. So we save money.

[01:13:50] Speaker 2: We'll see. We'll see. Sounds good.

[01:13:52] Speaker 1: So one or two papers. So there's the it's like so there's that.

[01:13:54] Speaker 2: Yeah.

[01:13:55] Speaker 1: But, uh, hopefully, yeah. I mean, you know what to do, right? You know what to do, right? Please tell me, you know, also. Yeah. The rest is, like, again, like, just if you're familiar with if you know the protocols that Rafael told you, just go with it. Just go with it.

[01:14:11] Speaker 2: Okay. Works.

[01:14:12] Speaker 1: Yeah. Just go with it. And then, like yeah. See, I mean and then involve other people, right? Like, again, like, if Vivi can just do and do, uh, plates and media while you do your biochemistry, that's, that's help, right? Awesome.

[01:14:35] Speaker 3: Uh, yeah. First of all, test, like, where usually do you like, like, Prof, where, like, where do you guys, like, find the protocol? Like, it's just from the, like, attached paper or, like, there's.

[01:14:47] Speaker 1: Yeah. It will be it will be the papers. It will be the papers. Yeah. Yeah. You can look at, uh, uh, uh, you can look at polymer D-polymerases, uh, spot test, spot assay, those, those keywords. I'm annoyed that this doesn't say how much they use. But let's see. Maybe based on, like, we could poten because they don't they don't even say what's the volume of culture that they have. Because based on that, we could potentially go back, right? Oh, hold on. Hold on. Actually, here it is. They did half a liter of LB for

their, uh.

[01:15:28] Speaker 2: Culture?

[01:15:29] Speaker 1: For the protein expression.

[01:15:30] Speaker 2: Mm-hmm.

[01:15:31] Speaker 1: And then and then they don't mention how much they how much they don't mention the concentrations. Uh, but they, they, they do say that, um like, whatever they alluded, they, they just start, like, just go with it. Just start, like, dropping it on top of the cultures. But we don't know how much they use.

[01:15:56] Speaker 2: Mm-hmm. All right. So here's what we're going to do later. Send you a drafted graph abstract. Figure out what part of the wiki who do.

[01:16:15] Speaker 1: Uh-huh.

[01:16:16] Speaker 2: And then figure out the timeline for following two weeks for the labs to do. And then for the labs, it's mainly into two different two different, like, sub-projects. The first one is about, uh, analyzing the already resistance bacterials genome. So we are going to if the PALA is not extracted to DNA genome, we are going to extract the genome and then do immuniprep, and then to send it for sequencing and analyze the sequence. That's the that's the first pass. The second pass is that, uh, we are going to keep doing the spot assay.

[01:16:52] Speaker 1: Hold on. Give me a second. Give me one second. Sorry.

[01:17:04] Speaker 2: Dial. I'm so sorry, guys. Give me one second.

[01:21:51] Speaker 3: Yeah. Yeah. Take your time.

[01:22:11] Speaker 1: I'm so sorry, guys. I'm so sorry, guys. Uh, yes. I'm so, so

sorry.

[01:22:15] Speaker 2: No worry. No worry. No worry. No worry. No worry.

[01:22:18] Speaker 1: Hopefully, you guys have enough stuff to do so you also did some stuff.

[01:22:23] Speaker 2: Yeah. Yeah. We're, we're, we're, we're discussing, uh, Vivi's and Vincent's schedule. We're figuring it out.

[01:22:32] Speaker 1: Awesome. Yeah. Yeah. Awesome. Yeah. Yeah. Yeah. You know, I mean, honestly, we can—you can use him to, like, uh, culture—uh, make media cultures. I'm pretty sure he's going to be happy, right? And then you guys focus on the important stuff, which is the biochemistry.

[01:22:48] Speaker 2: Yeah. Yeah. Sounds good. Oh, uh, so, so continue. We're left. So the other pass was doing the spot essay. First, we definitely need to we first need to have the protein cell. The extracted—extracted protein, which is, like, the purified protein, which is, like, and—and not purified protein. So, like.

[01:23:07] Speaker 1: That takes—that takes three days. That takes three days.

[01:23:09] Speaker 2: Yeah. Yeah. That takes three days.

[01:23:10] Speaker 1: And maybe you can start right away with, like, like, let's say worst-case scenario, like, you have enough you should have enough protein in the extraction that you already have, right?

[01:23:22] Speaker 3: We only keep one round, I think. We're—yeah. Because, like.

[01:23:26] Speaker 1: That's another thing. Like, why do we keep two illusions? Like, why do we keep the two all of these things are driving me insane.

[01:23:34] Speaker 3: You mean, like, why we do like, why we load gel for illusion one, two, and three.

[01:23:39] Speaker 1: Yes. Yeah. Yes. Yeah. Yeah.

[01:23:40] Speaker 3: That's what Rafael said. Rafael said.

[01:23:42] Speaker 1: I'm like, this is, like, driving me insane. Like, I know. I keep telling him, like, it's just the same. It's the same. It's the exact same thing.

[01:23:50] Speaker 3: Like, he said, like, maybe, like, like, through different illusions and, like, pre-wash and after-wash, like, the amount will be different.

[01:23:59] Speaker 1: Yeah. Yeah. Yeah.

[01:23:59] Speaker 3: So he wants.

[01:24:00] Speaker 1: I mean, if anything, let's, let's, let's, let's we learn something from this. We learn that whatever you did for E1 works, do it again. Maybe the problem is, like, you were juggling too many proteins at the same time, right? So let's just focus on one protein, right? Just, just do that one. And then, like, the illusions is just all of the illusions should be on a single—on a single tube, right? It's just—it's just protein, right? It's the same protein passing twice. So bless you.

[01:24:31] Speaker 2: Yeah. Yeah. Sounds good.

[01:24:33] Speaker 1: So, like, like, now you have that protein. That protein, you should be able to assay it right away, right? That might be, uh, uh, the E1, my hypothesis is that E1 is very similar to the wild type, so maybe not a big difference, right? So it should work the same way.

[01:24:55] Speaker 3: Okay. So. I'll—or we'll let Rafael know to keep, like, these rounds of protein soluble protein to keep that.

[01:25:05] Speaker 1: Yeah. Yeah. Yeah. Also, like, if you know how to do all of them, you don't need him anymore. There's that. Right?

[01:25:11] Speaker 3: But he'll still show up he'll still show up at the lab and, and sometimes to guide us. So, so, like.

[01:25:18] Speaker 1: I see. You can tell him you can tell him to, like, go write his papers. No, no. Because, I mean, the idea with Rafael was just, like, to get him, like, going, to get you guys going.

[01:25:33] Speaker 2: Yeah.

[01:25:33] Speaker 1: Right? But, like, if you know how if you know what to do, just do it. Right? And I'll ask Rafael to, like because he's also going he's also going to DC this weekend. He's going to ASN, to a conference. So, so, so just, so just I'll message him saying, like, you guys are good to go, and just go.

[01:25:55] Speaker 2: Okay. Sounds good.

[01:25:57] Speaker 3: Thanks for trusting us.

[01:26:00] Speaker 1: Yeah. No, I might because I want you guys to do it, like, ASAP, right? Because if you have to meet with Rafael and the schedule, it's complicated, right? And if you're on your own, you can do it, like, at 10:00 PM or at some time, right? Or wherever you guys want to do it, right?

[01:26:14] Speaker 2: Yeah.

[01:26:14] Speaker 1: I mean, don't stay late in the lab. Don't stay in the lab. But, like, I do understand when most undergrads are able to work best, like, 5:00 PM, 6:00 PM, right, after hours.

[01:26:25] Speaker 3: I do say you can this last, last Saturday. But it's because of gel. The machine is so I don't know what happened what's going on there. But, like. Just from, like, two hours to get the gel. I.

[01:26:38] Speaker 1: Oh, shoot. Yeah.

[01:26:39] Speaker 2: Cough.

[01:26:40] Speaker 3: I don't know what happened. Yeah.

[01:26:42] Speaker 2: Also, one last question. One last question other than the gel machine, which is, like, can we can we use the, like, proper, proper, like, gel picture thing in the lab next door? Because, like, I, I like using a phone to take pictures is, like, too hard. And then.

[01:27:00] Speaker 1: Yeah. Of course. But which imager, though?

[01:27:03] Speaker 2: The one we tried, but it doesn't work. And then.

[01:27:06] Speaker 1: I'll ask someone. I'll ask someone. That might actually mean we need someone then. Uh, because I think I can ask I can ask some students. I might ask, like, someone else.

[01:27:17] Speaker 2: Okay.

[01:27:18] Speaker 1: That's how to use it. Because I don't know how to use that.

[01:27:19] Speaker 2: Okay. Sounds good.

[01:27:20] Speaker 1: I really think, like I mean, it's okay. It's up to you guys. I really think, like, an iPad, bright, white background is more than enough. But yeah. Yeah.

[01:27:31] Speaker 2: Yeah.

[01:27:31] Speaker 1: I mean, like, if you guys want to and if you're going to go for it.

[01:27:34] Speaker 2: Okay. Yeah. Sounds good.

[01:27:37] Speaker 1: Yeah. Awesome. So it looks like you guys have a plan,

right? Like, start resurrecting I mean, if tomorrow you start growing a lot of the transformed bacteria. You should have protein by Thursday evening, right? If you have protein by Thursday evening. What's going on?

[01:27:58] Speaker 2: I'm thinking. I'm just thinking. Yeah.

[01:28:00] Speaker 1: If you have protein by Thursday evening, you should be able to, like you should have, like, bacteria already on a loan. So you can just, like, add your enzyme, right?

[01:28:13] Speaker 3: Okay.

[01:28:14] Speaker 2: Yeah. Technically. Technically.

[01:28:17] Speaker 1: Right?

[01:28:21] Speaker 3: We store our bacteria glycerol stock at minus 80, right?

[01:28:26] Speaker 2: Yes. Yes.

[01:28:26] Speaker 1: Yeah. Yeah. Yeah. The bacteria is on the steeper layer. Yes.

[01:28:30] Speaker 3: On the topmost level.

[01:28:32] Speaker 2: Yeah.

[01:28:32] Speaker 1: Yes. That's where you have your isolates, and that's where you have the transformed side of it.

[01:28:38] Speaker 2: Yeah. That's correct.

[01:28:42] Speaker 1: Yeah. Yeah. I know, I know it's a lot. I know it's a lot.

[01:28:48] Speaker 3: But we're going to use the raw bacteria, right? Like, the, the, the.

[01:28:51] Speaker 1: Yeah. The original.

[01:28:52] Speaker 2: The original.

[01:28:53] Speaker 1: The old, old ones. The OG.

[01:28:54] Speaker 3: Okay.

[01:28:55] Speaker 1: Yeah. The original isolates to test the, uh to test the host range and the, uh, the polymerase activity. Right?

[01:29:04] Speaker 3: Okay. Sounds good. Sounds good. We'll come up with a plan for the next two weeks.

[01:29:08] Speaker 1: Yeah.

[01:29:09] Speaker 3: And send it to you. And also for the, uh, graphical abstract, like, so.

[01:29:14] Speaker 1: Yeah. The only reason I want to see the graphical abstract is because I want to know that we're talking about the same thing.

[01:29:21] Speaker 2: Yeah.

[01:29:22] Speaker 3: So do we still need to send it? Like, is it handwritten, or do you want to show, like, an online format? Because I'm.

[01:29:28] Speaker 1: It has to be nice because you're going to it's one of your deliverables.

[01:29:31] Speaker 3: Oh.

[01:29:31] Speaker 1: Like.

[01:29:32] Speaker 3: Oh. I see. I see. Oh, I got it. Got it.

[01:29:35] Speaker 2: Got it.

[01:29:36] Speaker 1: But I want to see a draft or any form so I know we're talking about the exact same thing, right?

[01:29:41] Speaker 2: Yeah. Sounds good. Sounds good.

[01:29:44] Speaker 1: Yeah. And then, and then the question yeah. I mean, I think, I think, I think I think it's fast. I think it's good. Even if you, uh, yeah. No, I think, I think it's as long as you learn something from this whole, this whole thing. You have, you have a paper. You have, uh, you're going to have more papers. Uh, some of you are going to Paris, which is wonderful. Maybe the president of CMC can, uh, uh, sponsor something.

[01:30:10] Speaker 2: Yeah.

[01:30:11] Speaker 1: Share something.

[01:30:12] Speaker 2: Yeah.

[01:30:12] Speaker 1: I mean, as long as, like, yeah. I'm sure he's going to be very excited. So I'm glad, I'm glad you, you, you, uh, you managed to get a meeting. Yeah.

[01:30:22] Speaker 2: We'll see.

[01:30:24] Speaker 1: Awesome. Awesome. Thank you very much, guys. You are, like, you're awesome, honestly. I keep telling people that, uh, I have really good students. And it's amazing that you guys are undergrads, you know? Because I have, like I see other students, I'm like, "Oh my God." But you guys are, like, a different level. So that's.

[01:30:43] Speaker 2: Yeah. Very nice.

[01:30:44] Speaker 1: So thank you, guys.

[01:30:45] Speaker 2: Thank you, bro.

[01:30:45] Speaker 1: Thank you. Thank you.

[01:30:46] Speaker 2: Amazing growth.

[01:30:48] Speaker 1: Yeah. Yes. Yes. Yes. No. Hopefully, yeah. Yeah. Let's, let's get this done. And then, like, after you submit your idea, maybe you can take a week off if you want. And then we go back to work more on the actual paper, right?

[01:30:59] Speaker 2: Yeah.

[01:31:01] Speaker 1: Yeah.

[01:31:01] Speaker 2: We got this.

[01:31:02] Speaker 1: I'll think about it. Like, if you—I'll think about it. Like, maybe you want to submit your, uh, the face as another email? We can do that. And then we have an extra paper, right? Or, like, think about it. All of those things are, like, to think about. Awesome.

[01:31:19] Speaker 2: Awesome.

[01:31:21] Speaker 1: Okay. Well, thank you very much, guys. I'll let you go. Ciao. Ciao.

[01:31:24] Speaker 2: All right. Thank you, bro.

[01:31:26] Speaker 3: Bye.

[01:31:26] Speaker 2: Bye.

[01:31:26] Speaker 1: Thank you.

[01:31:27] Speaker 3: Thank you, prof.

[01:31:29] Speaker 1: No. Thank you. Bye.

[01:31:32] Speaker 3: How do you say it?

[01:31:33] Speaker 2: Like this. Um, I recorded the whole thing just now, and in a bit I'll summarize the meeting minutes. Hey, where's he?

[01:31:40] Speaker 3: I'm here. I'm here.

[01:31:41] Speaker 2: Oh, you're here.

[01:31:42] Speaker 3: I went with him. You two aren't together.

[01:31:44] Speaker 2: He's upstairs.

[01:31:47] Speaker 3: Oh, that's not it. No, I know what I need to do. I wrote it down, basically what the lab is supposed to do. But these two experiments specifically involve something that needs to start being checked tomorrow, and it hasn't been decided. So I think, about the timeline you mentioned, do you have anything later? If there's nothing else, then let's just get it done right now. Otherwise, otherwise, it'll be two and a half weeks.

[01:32:09] Speaker 2: I'll have a meal first. I haven't eaten yet. I'll eat first.

[01:32:12] Speaker 3: I haven't eaten either. I don't feel like eating anymore. I'm getting dizzy listening to this. And I'm not hungry.

[01:32:17] Speaker 2: How about we reconvene at 10:00? Reconvene at 10:00.

[01:32:25] Speaker 3: What? Like, restart this meeting?

[01:32:27] Speaker 2: Yeah.

[01:32:29] Speaker 3: 10:00. And today's goal is to list things out, what to do

over the next two weeks.

[01:32:33] Speaker 2: Right. Today's goal is to get the things the professor wants done. First, get the things the professor wants done.

[01:32:40] Speaker 3: Wait, what does he want? When is he supposed to get that graphical abstract?

[01:32:44] Speaker 2: I'll handle that. I'll handle that. Don't worry about that. Don't worry about that. I'll do it. I know what he's talking about.

[01:32:48] Speaker 3: And then Vivi and Vincent, if you don't want to bother these two, I can go ask them when they're free. But the condition is that we first lay out the plan for the next two weeks, and only then can I go see Vincent.

[01:33:03] Speaker 2: Okay. I can handle Vivi's side. I'll leave Vincent's side to you. I don't want to deal with Vincent. I don't want to deal with Vincent.

[01:33:09] Speaker 3: No, but do we need so many people to help us set things up?

[01:33:12] Speaker 2: We need help beyond just setting it up. For other experiments, I think one person's help would be better.

[01:33:19] Speaker 3: Then what is it? Right now we only have small tests. But we don't even know what to do with our own small tests.

[01:33:26] Speaker 2: Hey, we'll talk later. We'll talk later. We'll chat later. We'll talk later. Right now my attention is already at that critical point, you know? I'm very anxious right now.

[01:33:35] Speaker 3: I'm very anxious too. I'm really scared that we won't have anything. Like, if it's about small tests and nothing, nothing, nothing showing up. Like, what do we have now? Is that enough?

[01:33:48] Speaker 2: Man, living is so exhausting. Fuck.

[01:33:50] Speaker 3: I think if the small tests have something, then there's still something to talk about. If they don't, then what do we have to synthesize in our IGEN?

[01:34:01] Speaker 1: We'll see. We'll see. We'll see.

[01:34:04] Speaker 3: Just eat. Just eat. Bye-bye. See you at 10:00.

[01:34:07] Speaker 2: Later, later.

[01:34:07] Speaker 3: Still use that one? Or, or...

[01:34:09] Speaker 2: We'll talk then. At the time, I might message you directly on WeChat too.

[01:34:13] Speaker 3: Uh, better to have like a visual screen, you know?

[01:34:16] Speaker 2: Okay. Sure, sure, sure. Sure, sure, sure. No problem.

[01:34:19] Speaker 3: Oh, OK. At 10:00, call Oli in too.

[01:34:21] Speaker 2: Okay. No problem. Let's leave it at that for now.

[01:34:24] Speaker 3: See you at 10:00. Bye-bye.

[01:34:25] Speaker 2: Bye-bye. Bye-bye.